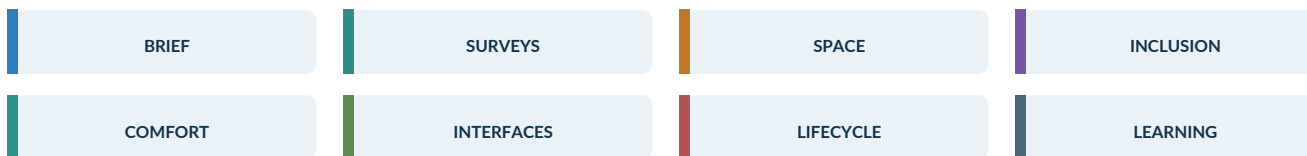
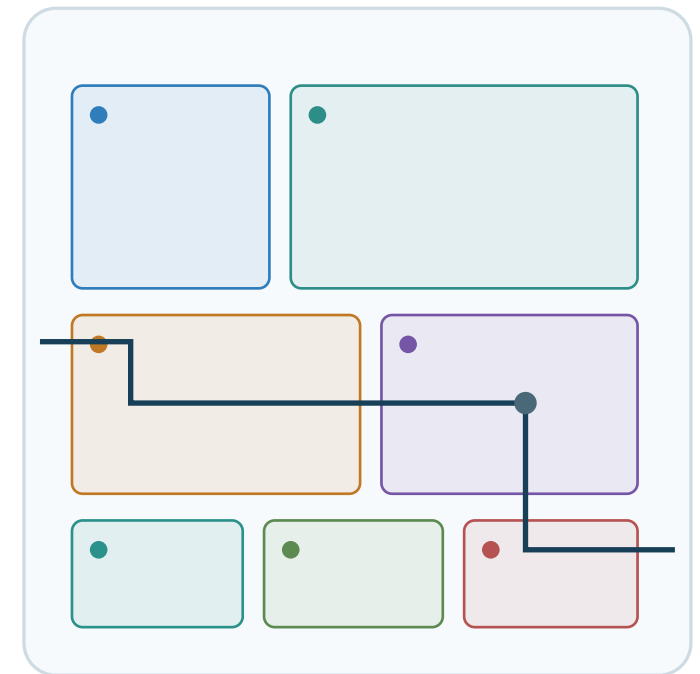


STAKEHOLDER 30

Architects and Building Designers

80 practical questions for evidence-led briefing, verified existing conditions, inclusive spatial design, coordinated technical interfaces, whole-life choices and post-occupancy learning.



Proposed sensing. Verified provenance. Explicit uncertainty. Accountable design action.

Scope and operating boundaries

PURPOSE

This catalogue helps architects and building designers frame answerable decisions, request proportionate evidence, assign authoritative owners and act without displacing professional judgement. It supports briefing, option appraisal, survey planning, coordination, commissioning and post-occupancy learning. It is not design authority, professional advice or a compliance certificate.

EVIDENCE STATUS AND TECHNICAL GATE

Every sensor, integration and analytic is proposed until deployment, asset and location mapping, commissioning, calibration or functional checking, permissions and current health are independently verified. Each use also needs clock and time-zone alignment, uptime or campaign completion, gap and drift checks, retained provenance and a documented aggregation path. A dashboard label is not proof of coverage, semantics, accuracy or fitness.

HISTORICAL SPATIAL REFERENCE

The May 2021 floor plan is historical spatial reference only. It cannot establish current room use, access, services, hazards, fire arrangements, ownership, availability or closures. Current controlled surveys, approved records, site verification and competent professionals always take precedence. Any conflict remains visible until the accountable owner resolves it.

NON-NEGOTIABLE PROFESSIONAL BOUNDARY

The service must not provide design sign-off; certify accessibility, safety, performance or statutory compliance; infer user needs without consultation; substitute telemetry for survey; authorise scope or work; override architects, engineers, clients, planning authorities, building control, dutyholders, responsible persons or system owners; or control building systems.

PRIVACY AND INCLUSION

Use the least intrusive evidence that can answer the decision. Do not identify or track people, inspect speech or work content, infer disability or protected characteristics, or expose restricted building information. Aggregate small groups and retain consultation evidence under its agreed consent and access conditions.

SAFE FAILURE

Insufficient evidence is a correct result. When authority, permission, physical coverage, calibration, timing, minimum evidence or professional competence is missing, stop the analytic claim, name the gap and assign the next authorised survey, consultation, calculation or decision.

USE EACH CARD TO MAKE A BOUNDED DECISION

Name the owner, interval, evidence gate, uncertainty, safe failure and next accountable action.

How to read each question card

1

STAKEHOLDER QUESTION

A natural architectural or building-design decision. It identifies the issue, decision point and accountable next action.

2

SENSORS AND TELEMETRY

Proposed modality, physical location, spatial and temporal coverage, sampling, aggregation, privacy and the full technical evidence gate.

3

AUTHORITATIVE SUPPORTING DATA

Named owners, controlled evidence, precedence, effective/observed/retrieved times, permissions, provenance, staleness and conflict route.

4

ANALYSIS REQUIRED

Operation type, decision interval, minimum evidence, method, uncertainty and safe failure. The result may be a bounded non-answer.

5

ANSWER BOUNDARY

What the service must not infer, disclose, promise, certify, authorise or control. Professional accountability stays human.

6

WHY THIS IS USEFUL

The concrete brief, survey, option, detail, specification, commissioning or learning action the stakeholder can take next.

OPERATION VOCABULARY

Lookup retrieves a controlled fact. Observation describes measured conditions. Calculation applies a stated formula. Estimate quantifies an unknown from evidence. Forecast projects a future range. Comparison tests alternatives or baselines. Diagnosis ranks testable explanations without claiming cause. Recommendation proposes an action only after authority and evidence gates pass.

TIME AND CONFLICT SEMANTICS

Effective time is when a requirement or approval applies. Observed time is when a condition was measured. Retrieved time is when the service obtained it. Recent retrieval does not make an old survey current, and recent telemetry does not override an approved requirement. Conflicts stay visible and go to the named owner.

Catalogue map: 80 questions in eight categories

1

Brief, stakeholders and user requirements

QUESTIONS 001-010

Freeze, change or assign requirements
Signed requirement status and unresolved owner

2

Existing conditions, surveys and constraints

QUESTIONS 011-020

Verify the base and commission targeted survey
Survey matrix, tolerances and gap commission

3

Spatial configuration, circulation and adjacencies

QUESTIONS 021-030

Select layouts, routes and operational relationships
Option comparison, route tests and fallback plan

4

Accessibility, inclusion and universal design

QUESTIONS 031-040

Remove barriers through consultation and measured evidence
Consultation-linked barrier actions and open issues

5

Daylight, lighting, acoustics, air and comfort

QUESTIONS 041-050

Coordinate environmental quality and cross-domain trade-offs
Performance criteria, trade-off record and test plan

6

Structure, services, interfaces and maintainability

QUESTIONS 051-060

Resolve technical interfaces and future access
Coordinated interfaces, access strategy and handover tests

7

Materials, lifecycle, adaptability and carbon

QUESTIONS 061-070

Choose whole-life, low-carbon and change-ready strategies
Carbon, lifecycle, reuse and adaptability comparison

8

Post-occupancy evidence, design validation and lessons

QUESTIONS 071-080

Compare outcomes and transfer bounded learning
Post-occupancy findings, confidence and future actions

DESIGN DECISION RHYTHM

Brief and option gates -> targeted survey and consultation -> coordination and mock-up -> commissioning -> seasonal post-occupancy assessment -> future standards and briefs

Brief, stakeholders and user requirements

AD-001

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which brief items can we freeze now, which remain measurable targets, and which assumptions need a named client decision before the next stage gate?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: No sensor determines the brief; supporting evidence may use anonymous threshold counts, room environmental loggers and owner-approved door-state events. Location and coverage: At room thresholds and representative occupied zones named in the current brief; supporting evidence only. Observation design: Representative teaching, research, event and quiet weeks after commissioning. 15-minute profiles and stage-gate summaries, never person-level traces. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No identity, device tracking or activity-content capture; telemetry cannot create, prioritise or approve a requirement and may represent only currently instrumented use.

AUTHORITATIVE SUPPORTING DATA

Owners: Client sponsor, project director and estates information owner. Controlled evidence: Signed brief, approved room data sheets, decision log and change-control register. Precedence: Signed and dated client decisions over workshop notes, models and telemetry. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the client sponsor and responsible designer.

ANALYSIS REQUIRED

Operation: Lookup + comparison. Decision task: Requirements traceability and unresolved-outcome classification. Decision interval: At each brief, concept, developed-design and pre-tender gate. Minimum evidence: Latest signed brief, named owner, decision status, rationale and representative evidence. Method: Map each outcome to fixed requirement, target, assumption or open decision; test evidence relevance. Uncertainty: Show confidence, evidence age and dissenting sources. Safe failure: Return unresolved and request owner decision rather than silently selecting an option.

ANSWER BOUNDARY

Do not infer individual needs, identities or protected characteristics; do not disclose restricted drawings, security details or person-level data; do not promise consensus, capacity, comfort or performance; do not certify design adequacy, accessibility, safety or statutory compliance; do not authorise scope, access, procurement, construction or operational changes; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Lock genuine client constraints, assign unresolved assumptions and trigger change control before the next design stage.

AD-002

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Who is materially affected by the design, whose evidence has changed the brief, and which missing voices require further consultation before we proceed?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: No sensor identifies stakeholder needs; supporting evidence may use anonymous entrance counts and accessible, consented feedback interfaces. Location and coverage: At public entrances and shared-facility approaches, plus accessible consultation terminals selected with the engagement lead. Observation design: During agreed consultation windows and representative operating periods. Weekly counts and theme totals with minimum-group thresholds. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No identity linkage, biometrics, free-text exposure or persistent tracking; counts show use, not representation, exclusion, disability or consent; non-participation can hide needs.

AUTHORITATIVE SUPPORTING DATA

Owners: Client engagement lead, equality and inclusion lead, students' and staff representatives. Controlled evidence: Consultation plan, attendance records, consented feedback, response logs and signed outcomes. Precedence: Direct, documented consultation with affected groups over inferred needs from telemetry. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the engagement lead and client sponsor.

ANALYSIS REQUIRED

Operation: Lookup + comparison. Decision task: Stakeholder coverage, influence and gap mapping. Decision interval: Before each design freeze and after material scope change. Minimum evidence: Stakeholder register, consultation method, response count, decision linkage and unresolved issue owner. Method: Compare affected groups with consulted groups; trace each material comment to accepted, rejected or pending decision. Uncertainty: Report sampling bias, inaccessible channels and missing voices. Safe failure: State insufficient representation and commission further consultation.

ANSWER BOUNDARY

Do not infer needs, disability, satisfaction or consent from observed behaviour; do not disclose restricted drawings, security details or person-level data; do not promise representation, agreement or acceptance; do not certify design adequacy, accessibility, safety or statutory compliance; do not authorise scope, access, procurement, construction or operational changes; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Commission missing engagement, adapt consultation methods and show which decisions are supported by which groups.

Brief, stakeholders and user requirements

AD-003

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which conflicts between user experience, operations, maintenance, security, cost and programme require an explicit client trade-off rather than a design-team assumption?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Overhead thermal or time-of-flight people counters, magnetic door contacts, owner-approved service timestamps, BMS or meter trends and calibrated environmental loggers where the trade-off requires them. Location and coverage: At the exact rooms, doors, service counters and plant branches implicated by the recorded trade-off. Observation design: Across at least two representative modes and any known peak period. Mode-tagged medians, percentiles and exception counts. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No individual route reconstruction or staff performance inference; correlation does not establish cause and operational workarounds can distort demand.

AUTHORITATIVE SUPPORTING DATA

Owners: Client sponsor and the accountable owner for each requirement. Controlled evidence: Brief clauses, operational procedures, maintenance strategies, security zoning and consultation outcomes. Precedence: Statutory duties and signed client requirements over preference statements; competent advice over analytics. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the project director through the formal decision process.

ANALYSIS REQUIRED

Operation: Comparison + recommendation. Decision task: Multi-criteria conflict analysis and decision-option comparison. Decision interval: Whenever a conflict affects layout, budget, programme or compliance. Minimum evidence: At least two credible options, named criteria, accountable owners, constraints and evidence for affected modes. Method: Construct a transparent trade-off matrix; test sensitivity to weights and assumptions. Uncertainty: Show non-comparable criteria, value judgements and option ranges. Safe failure: Escalate without recommendation when legal, safety or stakeholder authority is unresolved.

ANSWER BOUNDARY

Do not infer individual needs, identities or protected characteristics; do not disclose restricted drawings, security details or person-level data; do not promise a value-neutral optimum, universal satisfaction or risk elimination; do not certify design adequacy, accessibility, safety or statutory compliance; do not authorise trade-off acceptance or design change; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Select an explicit compromise with visible consequences, decision ownership and residual-risk ownership.

AD-004

ARCHITECTS AND BUILDING DESIGNERS QUESTION

What approved occupancy, coincident-arrival and dwell-time scenarios should size each room, threshold and shared amenity?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Commissioned overhead thermal or time-of-flight people counters, booking events, magnetic door contacts and CO2 only as a secondary occupancy proxy. Location and coverage: At every counted room entrance and selected shared-amenity threshold; CO2 probes sit in representative breathing zones away from doors and supply air. Observation design: 5-minute sampling over representative term, event and out-of-hours periods. 15-minute occupancy distributions, peak percentiles and dwell bands. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No face capture, device tracking or named attendance; counter overlap, open doors, CO2 lag, booking no-shows and atypical weeks reduce accuracy.

AUTHORITATIVE SUPPORTING DATA

Owners: Timetabling owner, room-booking owner, estates space manager and competent fire adviser. Controlled evidence: Current capacity records, approved use, timetable, booking policy and client growth assumptions. Precedence: Approved capacity and competent advice over sensor-derived estimates; current use owner over the May 2021 floor plan. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the space manager and responsible designer.

ANALYSIS REQUIRED

Operation: Observation + estimate. Decision task: Demand estimation and sizing-scenario comparison. Decision interval: At briefing, layout option assessment and pre-handover validation. Minimum evidence: Mapped room, at least four representative weeks or a justified shorter campaign, booking data and known closures. Method: Compare booked, observed and design occupancy; model percentiles and simultaneous demand. Uncertainty: Provide ranges for counting error, seasonality and future growth. Safe failure: Withhold a sizing conclusion where coverage or approved use is uncertain.

ANSWER BOUNDARY

Do not infer attendance, identity, productivity or purpose of any person; do not disclose person-level attendance or restricted booking details; do not promise consensus, capacity, comfort or performance; do not certify room capacity, escape capacity or code compliance; do not authorise scope, access, procurement, construction or operational changes; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Size rooms, toilets, circulation and shared resources against transparent demand rather than a single untested peak.

Brief, stakeholders and user requirements

AD-005

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which end-to-end user, teaching, research, cleaning, delivery, maintenance and emergency workflows must the plan support, and where are their critical hand-offs?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Overhead threshold counters, magnetic door contacts, owner-supplied lift-controller events, loading-bay loop or ultrasonic occupancy sensing and structured non-recording observation logs. Location and coverage: At agreed workflow hand-off points, loading thresholds, lift lobbies and service doors; manual observers remain stationary. Observation design: Sampled in representative shifts, changeovers and delivery windows. Task-stage durations and anonymous flow totals, not linked journeys. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No identity, audio, video retention or worker performance scoring; observed practice may differ from authorised procedure and rare emergency tasks may be absent.

AUTHORITATIVE SUPPORTING DATA

Owners: Service owner for each workflow, estates operations, security and safety leads. Controlled evidence: Current standard operating procedures, method statements, delivery rules, cleaning plans and emergency arrangements. Precedence: Authorised procedures and competent safety advice over observed shortcuts or historic drawings. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the relevant service owner and principal designer.

ANALYSIS REQUIRED

Operation: Observation + comparison. Decision task: Workflow decomposition, route-interface analysis and failure-point identification. Decision interval: During briefing and whenever layout or service strategy changes. Minimum evidence: Approved workflow, task owner interview, route geometry and at least one observed representative cycle where permitted. Method: Map actors, materials, equipment, decision points and hand-offs; compare normal and disrupted modes. Uncertainty: Label unobserved rare tasks and contested practices. Safe failure: Refer emergency or high-risk workflow questions to competent professionals.

ANSWER BOUNDARY

Do not infer worker intent, competence or productivity; do not disclose sensitive research, security or staff information; do not promise uninterrupted operations or emergency effectiveness; do not certify safe systems of work or emergency plans; do not authorise scope, access, procurement, construction or operational changes; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Protect required adjacencies, clearances, storage and service routes before they become costly operational workarounds.

AD-006

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which operating modes must the building support, and what layout, access or service-state changes are needed between them?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Mode-tagged overhead people counters, magnetic door contacts, owner-supplied lift and BMS point states, non-recording sound-level instruments, room environmental loggers and verified energy meters. Location and coverage: At mode-affected entrances, lift interfaces, representative room zones and relevant BMS points or meters. Observation design: Across representative examples of each mode after mapping and commissioning. 15-minute profiles plus mode start, peak and recovery summaries. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No participant identification or content capture; rare modes, manual overrides and incomplete mode labels can invalidate comparisons.

AUTHORITATIVE SUPPORTING DATA

Owners: Timetabling, events, security, estates operations and client sponsor. Controlled evidence: Approved calendars, event plans, access rules, closure notices and operating-mode definitions. Precedence: Current authorised mode records over inferred mode labels or historical plan annotations. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the accountable mode owner and project director.

ANALYSIS REQUIRED

Operation: Comparison + forecast. Decision task: Scenario requirement extraction and cross-mode stress testing. Decision interval: At brief approval, option selection and commissioning-plan definition. Minimum evidence: Named mode, responsible owner, occupancy range, access pattern, service state and disruption assumptions. Method: Build a mode matrix and test each design option against simultaneous spatial and service demands. Uncertainty: Show absent modes and uncertain coincidence assumptions. Safe failure: State not assessed where no authorised scenario exists.

ANSWER BOUNDARY

Do not infer purpose or status of individuals from access or occupancy; do not disclose restricted drawings, security details or person-level data; do not promise that every future event or closure can be accommodated; do not certify event, security or emergency approval; do not authorise scope, access, procurement, construction or operational changes; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Avoid a design that works only in a nominal weekday mode and define testable requirements for transitions and recovery.

Brief, stakeholders and user requirements

AD-007

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which privacy, dignity and confidentiality requirements should determine sightlines, acoustic separation, access transitions and data-enabled features?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Non-recording sound-level instruments, measured sightline studies, door-state events and consented privacy walkthroughs. Location and coverage: At consultation-approved privacy interfaces, doors and acoustic boundaries; sound instruments record levels only. Observation design: Short diagnostic campaigns and agreed post-occupancy checks. Zone-level exceedance counts and time bands. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No audio recording, image retention, content inspection, device identifiers or inferred sensitive activity; sound level does not establish speech intelligibility, dignity or consent, and a sightline survey does not predict every use.

AUTHORITATIVE SUPPORTING DATA

Owners: Client data owner, data-protection officer, equality lead, security lead and affected users. Controlled evidence: Privacy impact assessments, information-governance policy, room requirements, access policy and consultation outcomes. Precedence: Lawful, approved privacy requirements and direct consultation over behavioural inference. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the data-protection officer and responsible designer.

ANALYSIS REQUIRED

Operation: Observation + comparison. Decision task: Privacy-by-design threat modelling and requirement traceability. Decision interval: Before fixing layouts, glazing, acoustic separations or sensing specifications. Minimum evidence: Defined use case, data flow, affected groups, lawful basis, retention and human consultation. Method: Identify observation, disclosure, inference and misuse paths; apply data minimisation and separation. Uncertainty: State residual risk and assumptions about future technology. Safe failure: Reject the data-enabled feature when lawful, proportionate safeguards are not demonstrated.

ANSWER BOUNDARY

Do not infer confidential activity, health, belief, identity or consent; do not disclose private conversations, research content, access patterns or sensitive room use; do not promise complete confidentiality; do not certify legal compliance or acoustic privacy; do not authorise scope, access, procurement, construction or operational changes; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Revise views, thresholds, partitions and data features before privacy harm is embedded in the building.

AD-008

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Where should public, shared, controlled and restricted zones begin and end, and what must happen at each inclusive threshold?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Overhead thermal or time-of-flight threshold counters, magnetic door contacts, owner-supplied aggregate access outcome categories and assistance-call event logs. Location and coverage: At mapped transitions between public, shared, controlled and restricted zones; access logs remain in the owner system. Observation design: Representative hours plus planned event and out-of-hours modes. 15-minute demand, denied-access reason categories supplied by the owner, and exception counts. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No facial recognition, credential disclosure or linked movement profiles; access denials can reflect permissions rather than poor design; security events may be restricted.

AUTHORITATIVE SUPPORTING DATA

Owners: Security manager, access-control owner, client sponsor and inclusion lead. Controlled evidence: Current zoning policy, threat assessment, credential rules, reception procedures and accessible access requirements. Precedence: Security authority and statutory duties over convenience analytics; current controlled-zone map over the May 2021 plan. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the security manager and client sponsor.

ANALYSIS REQUIRED

Operation: Lookup + observation + comparison. Decision task: Zonal access, threshold-capacity and wayfinding trade-off analysis. Decision interval: At concept and developed design, and after security policy change. Minimum evidence: Approved zone model, user groups, assisted-access process, peak demand and alternative routes. Method: Model entry decisions, queue formation and route legibility without weakening required controls. Uncertainty: Separate policy-driven denial from spatial friction and show missing security information. Safe failure: Provide only bounded options when threat information or authority is unavailable.

ANSWER BOUNDARY

Do not infer threat, intent, clearance or protected status from movement; do not disclose security vulnerabilities, credential rules or restricted-zone details; do not promise security, inclusion or frictionless access; do not certify security adequacy or access compliance; do not authorise zone classification, credentials or security works; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Locate receptions, doors, waiting areas and signs so required control is understandable, accessible and operationally manageable.

AD-009

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which credible growth, technology and organisational scenarios should the design absorb without major structural or services work?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: No sensor predicts organisational change; use quality-controlled utilisation, movable-partition states, service headroom and asset-load histories as scenario inputs. Location and coverage: At adaptable-space thresholds, movable-partition states and metered plant or distribution assets serving those zones. Observation design: Monthly collection with annual planning snapshots. Quarterly percentiles, trend bands and change-event summaries. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No team or individual performance profiling; past trends do not establish future strategy, and changed boundaries or uncommissioned meters break comparability.

AUTHORITATIVE SUPPORTING DATA

Owners: Client sponsor, estates strategy owner, departmental leadership and ICT or laboratory service owners. Controlled evidence: Approved estate strategy, growth scenarios, technology roadmaps, lease constraints and asset plans. Precedence: Board-approved strategy and current service standards over extrapolated telemetry. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the client sponsor through portfolio governance.

ANALYSIS REQUIRED

Operation: Forecast. Decision task: Scenario planning, capacity-headroom and adaptation-cost comparison. Decision interval: At briefing, option selection and major scope change. Minimum evidence: At least three plausible demand scenarios, trigger points, service limits and adaptation assumptions. Method: Test loose-fit layouts, distribution capacity and staged interventions under low, central and high cases. Uncertainty: Show scenario dependence and non-linear thresholds. Safe failure: Avoid a preferred forecast; retain options where strategic uncertainty is material.

ANSWER BOUNDARY

Do not infer future staffing, research success or organisational decisions from current use; do not disclose restricted drawings, security details or person-level data; do not promise future-proofing or a fixed adaptation horizon; do not certify capacity or investment sufficiency; do not authorise scope, access, procurement, construction or operational changes; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Reserve sensible headroom, choose reversible details and avoid both premature oversizing and brittle minimum provision.

AD-010

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which design-intent criteria, baselines, tolerances and acceptance methods must be agreed before options are tested and specified?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: An outcome-specific plan using calibrated lux, sound, temperature, humidity, CO2, energy or water instruments and anonymous use counts only where each measure maps to a signed criterion. Location and coverage: At each space or asset tied to a signed outcome criterion, using a documented measurement position. Observation design: Baseline before change where possible, functional testing at handover, then seasonal checks. Criterion-specific statistics with raw-data retention governed separately. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: Anonymous feedback and non-identifying telemetry by default; poor baseline, seasonal mismatch and sensor replacement can make targets incomparable.

AUTHORITATIVE SUPPORTING DATA

Owners: Client sponsor, contract administrator, commissioning manager and each performance owner. Controlled evidence: Employer's requirements, performance specifications, design intent, commissioning plan and acceptance schedule. Precedence: Contractual acceptance criteria and competent test results over dashboard summaries. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the contract administrator and named criterion owner.

ANALYSIS REQUIRED

Operation: Calculation + recommendation. Decision task: Metric definition, evidence-plan validation and acceptance traceability. Decision interval: When requirements are written, before tender and before each acceptance test. Minimum evidence: Criterion, threshold or target, method, instrument, timing, owner, tolerance and remedy path. Method: Check that every criterion is specific, measurable, attributable and testable under stated conditions. Uncertainty: State measurement tolerance, confounders and unmet preconditions. Safe failure: Mark the criterion untestable rather than inventing a proxy.

ANSWER BOUNDARY

Do not infer satisfaction or design success from a single metric; do not disclose restricted commissioning or respondent data; do not promise acceptance or future performance; do not certify contract completion, compliance or professional sign-off; do not authorise payment, handover or remedial work; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Turn broad aspirations into acceptance criteria, evidence tasks, commissioning checks and post-occupancy tests.

Existing conditions, surveys and constraints

AD-011

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which geometry and attributes in the current base model are survey-verified, to what tolerance, and which remain inferred?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: A commissioned terrestrial laser scan, total-station control and targeted measured checks. Location and coverage: On a surveyed control network covering the intervention area, interfaces, elevations and access routes; scan stations avoid occupied capture. Observation design: Single dated survey after access clearance, with resurvey after material change. Registered point cloud, tolerance report and element-level confidence tags. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: Survey capture scheduled to avoid people, with incidental imagery removed or restricted; occlusion, reflective surfaces, inaccessible voids and registration error leave unknowns.

AUTHORITATIVE SUPPORTING DATA

Owners: Appointed measured-survey provider and estates information manager. Controlled evidence: Current controlled survey, verified BIM or CAD model, change records and the May 2021 floor plan as historical reference only. Precedence: New competent measured survey over legacy drawings and unverified models. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the survey lead and responsible designer.

ANALYSIS REQUIRED

Operation: Observation + comparison. Decision task: Geometric reconciliation and model-confidence classification. Decision interval: Before option coordination, setting-out and tender information. Minimum evidence: Survey control, tolerance, date, coverage map and model version. Method: Compare model geometry with surveyed points; classify verified, inferred, inaccessible and conflicting elements. Uncertainty: Report deviation ranges and unobserved areas. Safe failure: Do not coordinate critical work to unverified geometry; request targeted resurvey.

ANSWER BOUNDARY

Do not infer current use, ownership, hazards or services from geometry; do not disclose restricted survey imagery or secure-area geometry; do not promise complete hidden-condition accuracy; do not certify survey adequacy, title boundaries or construction setting-out; do not authorise scope, access, procurement, construction or operational changes; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Select a dependable base, mark tolerances and unknowns, and commission only the additional survey that the decision needs.

AD-012

ARCHITECTS AND BUILDING DESIGNERS QUESTION

What verified levels, slopes, floor-to-floor heights, build-ups and clear headrooms constrain the proposal?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Digital level, total station, laser distance measurement and targeted inclinometer monitoring where a competent engineer specifies. Location and coverage: At floors, ramps, landings, stairs, ceiling grids and known headroom pinch points affected by the proposal. Observation design: Dated survey before design freeze; monitoring only for a defined movement concern. Spot levels, slope bands and minimum-clearance maps. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No personal data; survey access records kept separately; finishes, ceiling deflection, clutter, inaccessible voids and instrument setup affect results.

AUTHORITATIVE SUPPORTING DATA

Owners: Measured surveyor, structural engineer and estates information owner. Controlled evidence: Controlled level survey, structural drawings, approved datum and verified reflected-ceiling information. Precedence: Competent current survey and agreed project datum over legacy dimensions. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the surveyor and responsible discipline lead.

ANALYSIS REQUIRED

Operation: Observation + calculation. Decision task: Clearance, gradient and datum-consistency analysis. Decision interval: Before fixing floors, ramps, services and ceiling build-ups. Minimum evidence: Verified datum, representative points at changes, survey tolerance and proposed build-ups. Method: Construct a level model; test minimum clearances and gradients with tolerance allowances. Uncertainty: Show measurement tolerance, movement allowance and unresolved datum conflicts. Safe failure: Return survey gap where critical clearance is not directly verified.

ANSWER BOUNDARY

Do not infer accessibility or structural movement from sparse levels; do not disclose restricted drawings, security details or person-level data; do not promise achievable clearance after construction tolerances; do not certify accessibility, structure or setting-out; do not authorise scope, access, procurement, construction or operational changes; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Coordinate thresholds, ramps, ceilings and service routes without relying on nominal dimensions that may not exist.

Existing conditions, surveys and constraints

AD-013

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which structural elements and load paths are confirmed, and what engineer-led investigation is needed before we place openings, fixings or new loads?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Engineer-specified cover meter, rebar scan, ground-penetrating radar, trial opening, deflection or strain monitoring. Location and coverage: Only on structural elements named by the structural engineer, with test grids and opening-up locations marked in the survey. Observation design: Targeted surveys before design commitment and monitored tests under an approved plan. Element-level findings with calibration, test location and confidence. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No unrelated imaging or building-wide scanning; reinforcement congestion, hidden remedial works, access limits and indirect sensing create ambiguity.

AUTHORITATIVE SUPPORTING DATA

Owners: Appointed structural engineer and estates structural-record owner. Controlled evidence: Verified structural drawings, calculations, inspection findings, intrusive survey results and change history. Precedence: Current structural engineer assessment over visual appearance, old drawings or generic sensor thresholds. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the structural engineer.

ANALYSIS REQUIRED

Operation: Observation + bounded diagnosis. Decision task: Structural constraint classification and option screening. Decision interval: Before locating openings, heavy equipment or façade changes. Minimum evidence: Identified element, load case, survey evidence, material assumption and structural engineer assessment. Method: Map confirmed, probable and unknown structure; compare options without calculating unauthorised capacity. Diagnosis means ranked, testable hypotheses rather than causal proof. Uncertainty: State unknown reinforcement, material properties and load-path uncertainty. Safe failure: Stop and require competent structural investigation for any safety-critical decision.

ANSWER BOUNDARY

Do not infer load capacity or reinforcement solely from indirect telemetry; do not disclose restricted structural-security details; do not promise that an opening or load is feasible; do not certify structural adequacy; do not authorise cutting, drilling, loading or temporary works; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Avoid impossible options early and give the engineer a precise, bounded investigation brief.

AD-014

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Where do existing mechanical, electrical, public-health and data services actually run, what spare capacity exists, and which access zones must remain clear?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Read-only BMS trends, clamp metering, thermal imaging, cable tracing, ultrasonic flow checks, borescope and targeted scan. Location and coverage: At affected risers, ceiling-void access points, plant interfaces, boards and pipe or cable branches under permit. Observation design: Short diagnostic campaigns under owner control plus representative peak trend periods. Asset or branch summaries, not network-content capture. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No packet inspection, credential access or unrelated room monitoring; unmapped branches, insulation, inaccessible voids, temporary loads and sensor scaling can obscure reality.

AUTHORITATIVE SUPPORTING DATA

Owners: Estates MEP owner, appointed engineers and system custodians. Controlled evidence: Current as-builts, O&M; manuals, schematics, point lists, test certificates and approved change records. Precedence: Competent survey and owner-verified live-system information over legacy drawings and inferred routes. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the relevant system owner and discipline engineer.

ANALYSIS REQUIRED

Operation: Observation + bounded diagnosis. Decision task: Service-route reconciliation, headroom screening and access-gap mapping. Decision interval: Before coordinated design and before intrusive opening-up. Minimum evidence: Verified service identity, route evidence, operating state, design duty and maintenance requirement. Method: Overlay surveyed and recorded routes; classify confirmed, inferred and unknown; compare observed demand with rated capacity. Diagnosis means ranked, testable hypotheses rather than causal proof. Uncertainty: Show sensor tolerance, peak uncertainty and inaccessible zones. Safe failure: Do not assume hidden continuity or spare capacity; request isolation and survey plan.

ANSWER BOUNDARY

Do not infer service identity, capacity or safety from temperature or current alone; do not disclose network topology, credentials or security-sensitive service routes; do not promise spare capacity or clash-free installation; do not certify MEP adequacy; do not authorise connection, isolation or opening-up; do not control live plant, breakers, valves or networks.

WHY THIS IS USEFUL

Coordinate realistic service zones, protect access and identify where opening-up or capacity studies are essential.

Existing conditions, surveys and constraints

AD-015

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which fabric defects, moisture routes or movement conditions could compromise new finishes, partitions or junctions?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Calibrated moisture meters, temperature and humidity loggers, thermal imaging, crack gauges and leak-event sensors. Location and coverage: At representative and suspect wall, roof, floor and junction locations, with reference points in apparently dry fabric. Observation design: Spot survey plus at least one weather-responsive campaign where moisture is plausible. Daily ranges, event traces and defect-location records. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No continuous monitoring of occupied private activity; surface readings, drying cycles, concealed leaks, salts and seasonal weather can confound diagnosis.

AUTHORITATIVE SUPPORTING DATA

Owners: Building surveyor, estates maintenance owner and relevant specialist. Controlled evidence: Condition survey, defect history, work orders, leak reports, warranty records and intrusive findings. Precedence: Competent diagnosis and current inspection over isolated sensor alerts or cosmetic appearance. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the building surveyor and defect owner.

ANALYSIS REQUIRED

Operation: Observation + bounded diagnosis. Decision task: Defect triage, moisture-path hypothesis testing and interface-risk mapping. Decision interval: Before specifying finishes and after any remediation. Minimum evidence: Location, construction build-up, calibrated measurements, weather or service context and inspection. Method: Compare spatial and temporal patterns with plausible sources; distinguish symptom from cause. Diagnosis means ranked, testable hypotheses rather than causal proof. Uncertainty: Rank hypotheses and show unobserved concealed conditions. Safe failure: State diagnosis unresolved and recommend targeted opening-up or specialist testing.

ANSWER BOUNDARY

Do not infer mould, structural cause or habitability from a single reading; do not disclose occupant complaints beyond need-to-know; do not promise dryness or defect elimination; do not certify fabric safety or remediation adequacy; do not authorise intrusive works or closure; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Avoid sealing in defects, detail robust interfaces and sequence remediation before new work.

AD-016

ARCHITECTS AND BUILDING DESIGNERS QUESTION

What is the currently verified arrangement of fire compartments, doorsets, penetrations and escape interfaces within the work area?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Owner-supplied door-position or hold-open events and alarm-interface states can test operation; compartment lines, doorsets and penetrations require direct competent inspection. Location and coverage: At fire doors and alarm interfaces within or serving the work area; compartment fabric requires direct competent inspection. Observation design: Continuous event logging after commissioning, summarised weekly; supplemented by competent inspection. Door exception counts and mapped inspection status. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No person tracking or disclosure of security-sensitive alarm detail; telemetry cannot verify fire resistance, seals, compartment continuity, evacuation capacity or code compliance.

AUTHORITATIVE SUPPORTING DATA

Owners: Responsible person, fire-safety manager, competent fire engineer and building-control records owner. Controlled evidence: Current fire strategy, fire-risk assessment, door and compartment surveys, certification and approved changes. Precedence: Current competent fire-safety assessment and statutory decisions over the May 2021 plan, sensor status or visual assumption. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the responsible person and fire engineer.

ANALYSIS REQUIRED

Operation: Lookup + observation + comparison. Decision task: Record reconciliation and design-interface risk screening. Decision interval: Before concept freeze, detailed coordination and completion inspection. Minimum evidence: Current strategy, mapped compartment line, competent survey status, affected penetrations and proposed changes. Method: Compare controlled records and observed interfaces; flag missing evidence, conflicts and design dependencies. Uncertainty: Show uninspected zones and document age. Safe failure: Do not answer adequacy; refer every conflict or change to the competent fire professional.

ANSWER BOUNDARY

Do not infer fire resistance, safe escape or compliance from door telemetry; do not disclose restricted alarm or security configuration; do not promise life safety or approval; do not certify fire strategy, compartmentation, escape or compliance; do not authorise fire-system changes, occupation or work; do not control fire doors, alarms or smoke-control systems.

WHY THIS IS USEFUL

Coordinate penetrations and layouts around verified fire constraints and commission the right competent assessment.

Existing conditions, surveys and constraints

AD-017

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Where are utility entries, meters and isolation boundaries, and is verified service headroom sufficient for the proposed demand?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Submeters, clamp meters, flow and pressure loggers, valve or breaker status and utility-quality monitoring. Location and coverage: At authorised incoming supplies, meters, main boards, plant headers and affected downstream branches. Observation design: Representative peak and low-demand periods, with event logging for interruptions. Half-hourly demand profiles, peak percentiles and data-quality flags. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No user-level consumption or unauthorised access to control systems; meter class, scaling, shared loads, undocumented branches and short campaigns limit headroom estimates.

AUTHORITATIVE SUPPORTING DATA

Owners: Utility provider, estates energy manager, MEP system owner and appointed engineer. Controlled evidence: Current utility agreements, single-line diagrams, schematics, meter register, isolation schedule and capacity studies. Precedence: Provider limits and engineer-verified records over dashboard estimates. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the relevant utility or system owner and discipline engineer.

ANALYSIS REQUIRED

Operation: Observation + calculation + estimate. Decision task: Utility-boundary mapping and demand-headroom screening. Decision interval: At brief, developed design and before connection applications. Minimum evidence: Verified meter mapping, design duty, representative demand history, diversity assumption and known future loads. Method: Normalise demand, compare peaks with rated limits and test scenarios. Uncertainty: Show meter error, unmetered load and coincidence uncertainty. Safe failure: State headroom unknown and require formal capacity assessment where mapping or duration is inadequate.

ANSWER BOUNDARY

Do not infer isolation safety or spare capacity from telemetry alone; do not disclose restricted utility or security information; do not promise connection availability; do not certify capacity, electrical safety or water quality; do not authorise utility application, isolation or connection; do not control breakers, valves or meters.

WHY THIS IS USEFUL

Decide whether to redesign loads, reserve space for upgrades or initiate provider enquiries early.

AD-018

ARCHITECTS AND BUILDING DESIGNERS QUESTION

What hazardous materials or contaminated conditions may be disturbed, and what specialist evidence is required before scope or method is fixed?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Accredited material sampling, specialist gas or vapour instruments where risk-assessed, and chain-of-custody laboratory results. Location and coverage: At specialist-defined sample points in planned disturbance zones, voids and suspect materials; no building-wide extrapolation. Observation design: Before disturbance and during authorised works as specified by competent specialists. Sample-level results and area status, not broad inferred labels. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: Restricted role access and no personal exposure profiling; negative samples do not clear unsampled material; hidden layers and changing conditions remain possible.

AUTHORITATIVE SUPPORTING DATA

Owners: Estates health-and-safety owner, asbestos dutyholder, competent surveyor and environmental specialist. Controlled evidence: Current asbestos register, refurbishment or demolition survey, hazardous-material records, spill history and laboratory certificates. Precedence: Statutory register and competent targeted survey over age-based inference, visual appearance or telemetry. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the dutyholder and competent specialist.

ANALYSIS REQUIRED

Operation: Lookup + observation. Decision task: Hazard-register query, disturbance-path screening and survey-gap analysis. Decision interval: Before specifying demolition, penetrations or opening-up. Minimum evidence: Exact work scope, current register, survey coverage, sample provenance and access constraints. Method: Intersect proposed disturbance with verified and unverified material zones; classify clear, controlled and unknown. Uncertainty: State sampling limits and inaccessible areas. Safe failure: Treat the area as unknown and stop design assumptions until competent survey evidence exists.

ANSWER BOUNDARY

Do not infer absence, type or safety of hazardous material from building age or sensors; do not disclose restricted hazard records beyond authorised users; do not promise a hazard-free area; do not certify clearance, exposure safety or legal compliance; do not authorise disturbance, entry or remediation; do not control ventilation or emergency systems.

WHY THIS IS USEFUL

Avoid unsafe disturbance, define survey extents and coordinate containment or redesign before tender.

Existing conditions, surveys and constraints

AD-019

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which current planning, conservation, title, boundary, easement or neighbour constraints materially limit the design options?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: No sensor establishes legal or planning rights; use a controlled boundary or topographic survey and only the noise, vibration, traffic, daylight or weather measurements required by an agreed assessment. Location and coverage: At legal-boundary survey points and only the external receptor positions required by an agreed planning or neighbour assessment. Observation design: Representative periods matched to the assessment methodology. Period summaries with instrument and weather quality flags. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: Avoid identifiable neighbour or occupant capture and restrict sensitive property information; measured context cannot resolve title, policy interpretation, consent, precedent or a planning authority decision.

AUTHORITATIVE SUPPORTING DATA

Owners: Planning authority, client legal adviser, landowner, conservation adviser and specialist consultants. Controlled evidence: Adopted policy, listed or conservation records, title and easement documents, approvals, covenants and consultation correspondence. Precedence: Formal legal and planning records plus competent advice over historic drawings or sensor evidence. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the client legal adviser and planning lead.

ANALYSIS REQUIRED

Operation: Lookup + observation + comparison. Decision task: Constraint-register compilation and consent-pathway mapping. Decision interval: During feasibility and after material design change. Minimum evidence: Current site boundary, applicable designation, legal document, policy version, proposal and consultation status. Method: Map each constraint to affected design element, evidence owner, consent route and programme risk. Uncertainty: Show interpretation dependencies and pending authority views. Safe failure: State legal or planning advice required rather than predicting approval.

ANSWER BOUNDARY

Do not infer legal rights, consent likelihood or neighbour acceptance; do not disclose privileged legal advice or private neighbour data; do not promise planning approval or absence of challenge; do not certify title, policy compliance or consent; do not authorise submission, agreement or boundary work; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Avoid developing options that conflict with non-designable constraints and plan evidence for the correct consent route.

AD-020

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which existing-condition assumptions still lack decision-grade evidence, and what targeted survey or opening-up should be commissioned before design lock-in?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: No continuous sensor resolves survey assumptions; use the assumptions register, coverage map, targeted measured survey and authorised opening-up. Location and coverage: At every unverified interface on the assumptions register, using targeted survey stations or authorised opening-up points. Observation design: Updated after each survey, access change and design freeze. Coverage percentages and critical-gap lists by discipline. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: Survey capture avoids people and restricted areas unless permission is explicit; inaccessible voids and unopened construction remain unknown; absence of evidence is not confirmation.

AUTHORITATIVE SUPPORTING DATA

Owners: Project survey manager, responsible designer and each discipline lead. Controlled evidence: Survey brief, access log, completed reports, limitation statements, requests for information and controlled model. Precedence: Competent discipline assessment of decision-critical evidence over raw coverage percentages. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the project survey manager and responsible designer.

ANALYSIS REQUIRED

Operation: Comparison + recommendation. Decision task: Evidence sufficiency and proportionate-survey prioritisation. Decision interval: At feasibility, before tender and before any irreversible work package. Minimum evidence: Decision to be made, consequence of error, current evidence, stated tolerance and access constraints. Method: Rank gaps by decision criticality, uncertainty and cost of late discovery; select non-intrusive then intrusive methods. Uncertainty: Show residual unknowns and confidence after proposed survey. Safe failure: Block the dependent decision where minimum evidence is unmet.

ANSWER BOUNDARY

Do not infer hidden conditions from nearby surveyed areas; do not disclose restricted drawings, security details or person-level data; do not promise complete certainty after survey; do not certify survey sufficiency for professional sign-off; do not authorise opening-up, access or expenditure; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Target surveys before design lock-in and carry residual unknowns transparently into risk, scope and contract information.

Spatial configuration, circulation and adjacencies

AD-021

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which room dimensions, furniture arrangements, storage and service positions best support the intended tasks at realistic occupancy?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Current measured survey or laser scan, overhead thermal or time-of-flight counts, tagged furniture-state survey, booking records and short-term environmental loggers. Location and coverage: At representative room entrances, work settings and mock-up zones; geometry is tied to surveyed room coordinates. Observation design: Multiple sessions across normal, peak and alternative layouts. Session-level occupancy, density, setup time and unusable-area estimates. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No identity, attendance or productivity analysis; comparator spaces may differ in culture, equipment and management; counts do not show task quality.

AUTHORITATIVE SUPPORTING DATA

Owners: Client brief owner, room data-sheet owner, activity lead and estates space manager. Controlled evidence: Approved activity requirements, equipment lists, furniture standards, capacity records and consultation outcomes. Precedence: Current approved activity and competent capacity advice over historical room labels or occupancy averages. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the responsible designer and activity owner.

ANALYSIS REQUIRED

Operation: Observation + calculation. Decision task: Activity-space fit and layout-option comparison. Decision interval: At room-data-sheet approval and each material layout change. Minimum evidence: Activity sequence, people range, equipment, accessibility clearances, storage and at least one tested layout. Method: Simulate or mock up activity zones; compare net usable area, conflicts and setup burden. Uncertainty: Show sensitivity to furniture, group size and unverified equipment dimensions. Safe failure: State not demonstrated and require mock-up or user test.

ANSWER BOUNDARY

Do not infer learning, research or work quality from density; do not disclose restricted drawings, security details or person-level data; do not promise suitability for every future activity; do not certify capacity, accessibility or safe operation; do not authorise scope, access, procurement, construction or operational changes; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Adjust dimensions, proportions, furniture and storage before the plan becomes fixed.

AD-022

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which adjacencies are mandatory, preferred or incompatible, and what authorised workflow evidence justifies each ranking?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: No sensor sets adjacency priority; use workflow-owner consultation supported by anonymous threshold counts and structured non-recording observations. Location and coverage: At thresholds connecting candidate adjacencies and at observed hand-off points agreed with workflow owners. Observation design: Representative changeovers, deliveries and routine service periods. Transition totals and task-stage delays without linked person journeys. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No linked journeys, identity, work-content capture or staff performance scoring; observed frequency does not establish criticality, confidentiality, safety or future organisational need.

AUTHORITATIVE SUPPORTING DATA

Owners: Activity owners, operations leads, client brief owner and security or safety advisers where relevant. Controlled evidence: Approved workflow maps, adjacency requirements, risk controls, consultation records and room data sheets. Precedence: Authorised workflow and direct stakeholder evidence over raw transition frequency. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the responsible designer and requirement owner.

ANALYSIS REQUIRED

Operation: Comparison + recommendation. Decision task: Adjacency matrix construction and conflict-weighted option testing. Decision interval: During concept option development and before layout freeze. Minimum evidence: Named activity pairs, direction, frequency, consequence, privacy, security and service constraints. Method: Score essential, beneficial, neutral and incompatible relationships; test alternative block plans. Uncertainty: Show weight sensitivity and missing stakeholder evidence. Safe failure: Leave the relationship unresolved where owner authority or workflow evidence is absent.

ANSWER BOUNDARY

Do not infer collaboration quality, hierarchy or purpose from movement; do not disclose sensitive workflow or restricted-area information; do not promise better collaboration or efficiency; do not certify safe or compliant adjacency; do not authorise scope, access, procurement, construction or operational changes; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Prioritise scarce frontage and central locations for relationships that materially affect operation.

Spatial configuration, circulation and adjacencies

AD-023

ARCHITECTS AND BUILDING DESIGNERS QUESTION

At which locations will ordinary peaks, changeovers or events create queues, counterflows or waiting that spill into circulation?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Overhead time-of-flight or thermal people counters, door states and anonymous queue-depth sensing. Location and coverage: Overhead at selected corridor, stair, lobby and doorway pinch points; sensing fields stop before sensitive rooms. Observation design: 5-second or 1-minute sensing during representative peaks and approved event scenarios. 1-minute flow, density and queue percentiles with minimum-cell suppression. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No video retention, face analysis or cross-camera tracking; occlusion, counter overlap, groups, temporary furniture and rare surges can bias results.

AUTHORITATIVE SUPPORTING DATA

Owners: Estates operations, event owner, timetabling owner and competent fire or access advisers. Controlled evidence: Approved occupancy scenarios, event plans, circulation drawings, closure records and capacity constraints. Precedence: Competent life-safety and accessibility decisions over flow analytics; current authorised scenario over historic use. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the responsible designer and scenario owner.

ANALYSIS REQUIRED

Operation: Observation + estimate. Decision task: Pedestrian-flow screening and queue scenario simulation. Decision interval: At concept option assessment and before high-demand operation. Minimum evidence: Verified geometry, door state, arrival profile, route choice assumptions and at least one representative observed peak where available. Method: Calculate flow and density distributions; simulate credible surges and blocked elements. Uncertainty: Report model sensitivity, behavioural assumptions and unobserved extremes. Safe failure: Do not state safe capacity; refer critical findings for competent assessment.

ANSWER BOUNDARY

Do not infer identity, purpose or behaviour of individuals; do not disclose security-sensitive crowd patterns; do not promise queue elimination or safe egress; do not certify circulation, fire or crowd safety; do not authorise occupancy or event operation; do not control doors, barriers or lifts.

WHY THIS IS USEFUL

Widen, redistribute or sequence thresholds and provide waiting space before congestion is built in.

AD-024

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which stair, lift and lobby arrangement best serves routine peaks, inclusive access, deliveries and credible outage conditions?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Infrared beam or time-of-flight stair counters and owner-supplied lift-controller call, travel, door, fault and aggregate load-band data. Location and coverage: At stair entries, lift lobbies and approved lift-controller points; no cabin imaging or credential capture. Observation design: Continuous after commissioning with focused peak and outage observations. 5-minute demand and response percentiles by route and operating mode. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No camera, credential or person-level trip linkage; lift load sensors, grouped arrivals, maintenance modes and route substitution can distort demand.

AUTHORITATIVE SUPPORTING DATA

Owners: Lift system owner, estates operations, access adviser, delivery owner and competent fire adviser. Controlled evidence: Current lift specifications, maintenance records, access strategy, delivery rules and approved occupancy scenarios. Precedence: Competent access, lift and fire advice over telemetry; verified equipment specification over inferred capacity. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the responsible discipline leads.

ANALYSIS REQUIRED

Operation: Observation + estimate + forecast. Decision task: Vertical-demand, response-time and resilience scenario comparison. Decision interval: During core planning and before changes to lift or stair provision. Minimum evidence: Route geometry, lift performance data, demand profiles, mobility and delivery scenarios, and outage assumptions. Method: Compare normal and degraded modes; test waiting, transfer and service-route conflicts. Uncertainty: Show demand variability, unavailable user evidence and model limits. Safe failure: Do not recommend reduction where accessibility or safety authority is unresolved.

ANSWER BOUNDARY

Do not infer disability, destination or identity from lift use; do not disclose access-control or restricted movement data; do not promise waiting-time targets in all conditions; do not certify lift capacity, accessibility or evacuation safety; do not authorise lift use policy or occupancy; do not control lifts or doors.

WHY THIS IS USEFUL

Select core locations, waiting areas, service separation and redundancy requirements using bounded evidence.

Spatial configuration, circulation and adjacencies

AD-025

ARCHITECTS AND BUILDING DESIGNERS QUESTION

At which route decisions do first-time users hesitate or choose incorrectly, and what spatial or information change would remove that ambiguity?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Consented task-based wayfinding trials, timestamped observer notes, measured sign visibility or lighting and anonymous destination counts. Location and coverage: At nominated decision points and destinations during consented route trials; no persistent participant tracking. Observation design: Representative normal periods and controlled trials of alternative signs. Choice proportions, hesitation bands and assistance-request totals. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No persistent participant tracking, biometric capture or disclosure of individual difficulty; small samples, familiarity, language and temporary signs limit generalisation; hesitation does not reveal its cause.

AUTHORITATIVE SUPPORTING DATA

Owners: Client wayfinding owner, access adviser, security and affected user representatives. Controlled evidence: Current destination hierarchy, naming convention, sign standards, access routes and consultation evidence. Precedence: Approved destination and access information plus direct user testing over inferred behaviour. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the responsible designer and wayfinding owner.

ANALYSIS REQUIRED

Operation: Observation + comparison. Decision task: Decision-point legibility evaluation and wayfinding-option testing. Decision interval: At concept, signage design and after route or naming change. Minimum evidence: Destination set, user tasks, route options, sign prototypes and diverse participant testing. Method: Task-based walk-through, route-graph assessment and comparison of errors or assistance needs. Uncertainty: Report participant coverage, learning effects and temporary-condition gaps. Safe failure: State that legibility is not demonstrated and schedule further inclusive testing.

ANSWER BOUNDARY

Do not infer cognitive ability, language or disability from hesitation; do not disclose individual route-test results or secure destinations; do not promise that nobody will become lost; do not certify accessibility, safety or wayfinding compliance; do not authorise destination naming or security disclosure; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Simplify junctions, improve sightlines and coordinate naming, landmarks and signs before fabrication.

AD-026

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which delivery, waste, cleaning and maintenance routes avoid conflict with public movement, sensitive activities and protected finishes?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Loading-bay loop or ultrasonic occupancy sensors, magnetic service-door contacts, owner-supplied lift events, manual trolley counts and delivery-booking timestamps. Location and coverage: At loading bays, service doors, lift lobbies, corridor pinch points and destination thresholds during supervised trials. Observation design: Representative delivery and collection windows plus peak public periods. Job-level timings without worker identity and 15-minute route-demand summaries. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No tracking of workers, suppliers or research materials; unscheduled deliveries, manual propping and exceptional loads may be missed.

AUTHORITATIVE SUPPORTING DATA

Owners: Estates logistics, waste, cleaning, maintenance and security owners. Controlled evidence: Delivery rules, waste plans, cleaning schedules, asset replacement routes, access restrictions and method statements. Precedence: Authorised service procedures and safety requirements over observed shortcuts. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the relevant service owners and responsible designer.

ANALYSIS REQUIRED

Operation: Observation + comparison. Decision task: Service-route conflict, turning-clearance and time-window analysis. Decision interval: During layout development and construction-phasing planning. Minimum evidence: Vehicle or trolley envelope, load, route dimensions, access permissions, dwell time and sensitive periods. Method: Map route conflicts; test geometric clearances and schedule separation under normal and disrupted modes. Uncertainty: Show unknown load sizes and rare replacement tasks. Safe failure: Require a physical route trial or specialist assessment where clearance is marginal.

ANSWER BOUNDARY

Do not infer worker productivity or contents of deliveries; do not disclose supplier, security or sensitive-material details; do not promise conflict-free logistics; do not certify manual handling, traffic or operational safety; do not authorise delivery access or route use; do not control gates, doors or lifts.

WHY THIS IS USEFUL

Locate stores, waste rooms, loading points and service lifts so operations do not depend on unsafe or disruptive shortcuts.

Spatial configuration, circulation and adjacencies

AD-027

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which zoning and construction strategy best separates quiet, collaborative, noisy and confidential activities across operating modes?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Class 1 or suitable sound-level meters without audio recording, vibration sensors, occupancy bands and door states. Location and coverage: At boundaries between quiet, collaborative and noisy zones, with paired positions on both sides and occupancy context. Observation design: Short-interval logging across teaching, research, event and quiet periods. LAeq, percentile levels, event counts and co-occupancy summaries. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No speech recording, intelligibility inference or individual attribution; sound levels do not identify source, and short campaigns may miss intermittent equipment.

AUTHORITATIVE SUPPORTING DATA

Owners: Activity owners, acoustic consultant, timetabling owner and client brief owner. Controlled evidence: Room acoustic criteria, equipment schedules, operating modes, adjacency requirements and complaints history. Precedence: Competent acoustic design and approved use over unverified telemetry or historical room labels. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the acoustic consultant and responsible designer.

ANALYSIS REQUIRED

Operation: Observation + comparison. Decision task: Source-path-receiver analysis and acoustic-zoning option comparison. Decision interval: During block planning, partition design and after material use change. Minimum evidence: Source spectra or representative tests, partition paths, door states, occupancy mode and target criteria. Method: Compare measured or predicted transmission, flanking and schedule overlap for each option. Uncertainty: Show source uncertainty, future equipment assumptions and measurement limits. Safe failure: State further acoustic testing or consultant design is required.

ANSWER BOUNDARY

Do not infer conversation content, activity sensitivity or responsibility for noise; do not disclose complainant identity or research activity; do not promise silence, privacy or absence of complaints; do not certify acoustic privacy or compliance; do not authorise scope, access, procurement, construction or operational changes; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Relocate activities, strengthen separations, control doors or use scheduling as an explicit design dependency.

AD-028

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which spaces should reconfigure, how quickly, and what acoustic, service, accessibility, security and escape constraints must remain intact?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Magnetic or reed contacts on movable partitions, tagged furniture-state surveys, service-connector status, overhead anonymous counts and manual setup timing. Location and coverage: At movable partitions, service-connection panels, doors and surveyed clear zones used for reconfiguration trials. Observation design: At each trial configuration and periodically after occupation. Configuration-level setup time, faults, unmet-clearance and performance summaries. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No worker identity or task-performance scoring; sensors cannot verify fire, acoustic or access performance; undocumented furniture changes create mapping errors.

AUTHORITATIVE SUPPORTING DATA

Owners: Client brief owner, responsible designer, fire, access, acoustic and MEP discipline leads. Controlled evidence: Approved configuration drawings, room data sheets, operating instructions, component certification and change-control records. Precedence: Approved tested configurations and competent advice over telemetry or user-created layouts. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the responsible designer and relevant duty owner.

ANALYSIS REQUIRED

Operation: Observation + calculation + comparison. Decision task: Configuration rule checking and adaptation-friction comparison. Decision interval: Before design freeze, mock-up assessment and after repeated post-occupancy changes. Minimum evidence: Each intended configuration, component locations, clearances, services, door states and approved constraints. Method: Compare setup sequence, dependency conflicts and recoverability; maintain a configuration assurance matrix. Uncertainty: Show untested variants and component-tolerance uncertainty. Safe failure: Reject or label any unassessed configuration as unsupported.

ANSWER BOUNDARY

Do not infer that an observed layout is approved or safe; do not disclose restricted drawings, security details or person-level data; do not promise instant flexibility or equal performance in every configuration; do not certify fire, acoustic, access or operational adequacy; do not authorise reconfiguration or change of use; do not control partitions, doors or services.

WHY THIS IS USEFUL

Design genuine adaptability with clear limits rather than relying on movable elements that create hidden conflicts.

Spatial configuration, circulation and adjacencies

AD-029

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Where should shared toilets, lockers, kitchens, print points and informal settings be located and sized relative to verified demand?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Overhead thermal or time-of-flight counts at amenity lobbies, approved appliance or fixture run-state data, structured queue observations, fault logs and consumable weight or level sensors. Location and coverage: At amenity-lobby thresholds and replenishment points, never inside toilet cubicles or other private enclosures. Observation design: Continuous or sampled use across representative weekdays, peaks and events. 15-minute demand, queue and outage summaries with threshold suppression. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No identification or inference of personal activity within amenities; sensor failure, shared doors, blocked fixtures and unusual events can misstate demand.

AUTHORITATIVE SUPPORTING DATA

Owners: Estates service owner, client brief owner, access adviser and cleaning or maintenance owner. Controlled evidence: Approved provision standards, fixture inventory, service levels, access requirements, work orders and consultation evidence. Precedence: Statutory or approved provision and direct consultation over demand analytics. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the responsible designer and service owner.

ANALYSIS REQUIRED

Operation: Observation + calculation. Decision task: Catchment, demand-balance and outage-resilience analysis. Decision interval: During layout option assessment and after significant occupancy change. Minimum evidence: Resource capacity, opening hours, route distances, demand profile, accessible provision and outage history. Method: Map service catchments; compare peak demand and alternative provision when a cluster is unavailable. Uncertainty: Show counting error, unobserved unmet demand and policy assumptions. Safe failure: Do not recommend reductions where inclusive or statutory evidence is missing.

ANSWER BOUNDARY

Do not infer health, gender, disability or personal behaviour from amenity use; do not disclose person-level or sensitive amenity data; do not promise zero queues or equal experience; do not certify sanitary, accessibility or welfare compliance; do not authorise closure or provision change; do not control access or fixtures.

WHY THIS IS USEFUL

Relocate, resize or duplicate shared resources and plan resilient alternatives.

AD-030

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which fallback layouts and routes keep the plan workable when an entrance, corridor, lift or shared space is temporarily unavailable?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Magnetic door contacts, owner-supplied lift availability, operator-entered barrier states, overhead anonymous route counts and temperature or illuminance loggers on diversion routes. Location and coverage: At assets and thresholds used by each fallback route, with temporary closure states entered by authorised operators. Observation design: During planned closures or controlled tabletop and digital scenarios. Mode-specific availability and rerouted-demand summaries. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No individual route linkage or emergency tracking; simulated route choice may differ from real behaviour, and rare compound failures remain uncertain.

AUTHORITATIVE SUPPORTING DATA

Owners: Estates operations, access lead, security, event owner and competent fire adviser. Controlled evidence: Closure procedures, temporary access plans, approved alternative routes and occupancy scenarios. Precedence: Authorised closure and life-safety arrangements over analytics or historic plan assumptions. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the responsible operations and design leads.

ANALYSIS REQUIRED

Operation: Forecast + comparison. Decision task: Spatial resilience and degraded-mode scenario testing. Decision interval: At concept, phasing design and before planned closures. Minimum evidence: Verified route graph, closure scenario, affected groups, alternative capacity, signage and service dependencies. Method: Remove unavailable nodes and test route length, congestion, assistance and service continuity. Uncertainty: Show behavioural assumptions and unassessed combinations. Safe failure: Do not state safe operation; escalate scenarios lacking approved alternatives.

ANSWER BOUNDARY

Do not infer individual vulnerability or destination; do not disclose security-sensitive closure or route details; do not promise continued access in every disruption; do not certify emergency, fire or accessibility adequacy; do not authorise closure, occupancy or temporary route; do not control doors, lifts or barriers.

WHY THIS IS USEFUL

Build in alternative routes, sensible compartmentalisation and temporary-wayfinding provisions.

Accessibility, inclusion and universal design

AD-031

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Does every required journey form a continuous step-free route from arrival to destinations, amenities, assistance and departure?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Measured route survey, door and lift functional states, and consented walkthroughs with people who have relevant lived experience. Location and coverage: Along each relevant entrance-to-destination route at gradients, crossfalls, thresholds, doors, lifts and assistance points. Observation design: Survey at design stages; availability monitored by operating mode after commissioning. Route-level availability, gradient and obstruction summaries. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No disability inference, individual route publication or covert tracking; measurements cannot establish equitable experience, fatigue, dignity or suitability for every person.

AUTHORITATIVE SUPPORTING DATA

Owners: Access consultant, equality lead, estates operations and affected user representatives. Controlled evidence: Current access assessment, approved route drawings, lift records, door schedules, consultation outcomes and temporary-access plans. Precedence: Competent inclusive-design advice and direct consultation over geometric or telemetry inference. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the responsible designer and access lead.

ANALYSIS REQUIRED

Operation: Lookup + observation + calculation. Decision task: Accessible route-graph validation and degraded-mode comparison. Decision interval: At each layout freeze and when access routes change. Minimum evidence: Verified levels, widths, door operation, lift status, resting points, destination list and user consultation. Method: Trace continuous routes; identify dependencies, detours and single points of failure. Uncertainty: Show unverified surfaces, assistance assumptions and temporary-condition gaps. Safe failure: State route not demonstrated and commission access assessment or user test.

ANSWER BOUNDARY

Do not infer disability, independence or need for assistance; do not disclose individual access requirements without consent; do not promise independent access for every person; do not certify accessibility or legal compliance; do not authorise route designation, closure or occupation; do not control doors or lifts.

WHY THIS IS USEFUL

Remove route breaks, reduce reliance on staff assistance and define credible alternatives where continuity cannot be achieved.

AD-032

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which door leaf, threshold, force, ironmongery and hold-open choices provide inclusive use without creating security or fire-door conflicts?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Calibrated door-force gauge surveys, threshold laser measurement, automatic-door cycles, faults and opening-time telemetry. Location and coverage: At each sampled door leaf and approach zone, using measured clear-opening, threshold and force-test positions. Observation design: Survey before detailed specification and functional checks at commissioning and seasonal assessment. Door-level force, clearance, response and fault summaries. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No credential or user identity capture; wind, seals, closer adjustment, floor finish and battery condition change performance.

AUTHORITATIVE SUPPORTING DATA

Owners: Access consultant, door-system owner, fire-safety manager and responsible designer. Controlled evidence: Door schedule, hardware specification, access assessment, fire strategy, commissioning records and maintenance history. Precedence: Competent access and fire requirements plus measured tests over nominal product data. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the access consultant and relevant system owner.

ANALYSIS REQUIRED

Operation: Observation + comparison. Decision task: Door usability, interface and fault-pattern evaluation. Decision interval: At specification, mock-up, commissioning and after repeated complaints. Minimum evidence: Verified clear width, force profile, threshold, controls, fire requirement and representative operating conditions. Method: Compare measured performance with project criteria and user-test findings; isolate adjustment versus design issues. Uncertainty: Show environmental variation and untested user diversity. Safe failure: Do not pass the door; request competent adjustment, redesign or inclusive test.

ANSWER BOUNDARY

Do not infer strength, mobility or disability of users; do not disclose credential or assistance records; do not promise universal ease of use; do not certify fire-door, accessibility or product compliance; do not authorise door adjustment, hold-open use or access change; do not control door operators or locks.

WHY THIS IS USEFUL

Change leaf size, automation, ironmongery, lobby geometry or maintenance requirements before handover.

Accessibility, inclusion and universal design

AD-033

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which lift location, cabin, controls, waiting space, reliability and fallback arrangement best serves people and activities that depend on it?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Owner-supplied lift-controller uptime, call, travel, door, fault and aggregate load-band data, timed functional tests, measured car and control geometry, and lobby light or temperature checks. Location and coverage: At relevant lift lobbies, car controls and owner-approved controller points; dimensions are measured directly. Observation design: Continuous operational telemetry with monthly summaries and targeted user testing. Availability, wait-time and fault distributions by operating mode. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No CCTV, credential, destination or user identity linkage; fault codes do not show user experience; maintenance modes and rare entrapments require separate records.

AUTHORITATIVE SUPPORTING DATA

Owners: Lift owner, maintenance provider, access consultant, responsible person and user representatives. Controlled evidence: Lift specification, maintenance contract, statutory examination, access strategy, emergency procedure and consultation evidence. Precedence: Competent lift and access assessment over telemetry dashboards. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the lift owner and access lead.

ANALYSIS REQUIRED

Operation: Observation + comparison + estimate. Decision task: Lift dependency, usability and outage-resilience analysis. Decision interval: During core planning, commissioning and post-occupancy assessment. Minimum evidence: Lift dimensions and controls, route catchment, demand, fault history, alternative route and diverse user testing. Method: Compare dependency, wait, interface clarity and recovery under outages. Uncertainty: Show rare-event uncertainty and untested access needs. Safe failure: State that inclusive usability is not demonstrated and retain or provide alternatives.

ANSWER BOUNDARY

Do not infer disability or destination from lift use; do not disclose alarm-call or credential details; do not promise availability or response time; do not certify lift safety, accessibility or statutory compliance; do not authorise lift operation, evacuation use or closure; do not control lift movement or doors.

WHY THIS IS USEFUL

Refine core location, lobby, cabin, controls, redundancy and accessible alternatives.

AD-034

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Are accessible toilets, changing facilities and related amenities located, sized and equipped for the agreed users and operating modes?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Measured facility survey, door and assistance-call functional tests, fixture inventory and anonymous counts at the external lobby only where approved. Location and coverage: At the approach, lobby, door and fixture layout of each relevant facility; no occupancy sensing inside the room. Observation design: Routine service monitoring plus scheduled inspection and consultation. Facility-level availability, outage and service summaries. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No sensing inside the facility, no identity linkage and no inference of disability or personal care needs; use counts cannot establish adequacy, dignity, demand by user group or compliance.

AUTHORITATIVE SUPPORTING DATA

Owners: Access consultant, equality lead, estates service owner and affected user representatives. Controlled evidence: Approved room data sheets, measured surveys, equipment schedules, access assessment, cleaning and alarm-test records. Precedence: Competent inclusive-design advice and direct consultation over telemetry or nominal labels. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the access consultant and service owner.

ANALYSIS REQUIRED

Operation: Lookup + observation + comparison. Decision task: Facility provision, route, equipment and resilience assessment. Decision interval: At briefing, detailed design, commissioning and after service changes. Minimum evidence: Intended users, location, dimensions, equipment, alarm, transfer space, opening hours and alternative provision. Method: Check spatial and operational requirements; test route dependency and outage alternatives. Uncertainty: Show consultation limits and unverified equipment performance. Safe failure: State not demonstrated and require competent access assessment and user testing.

ANSWER BOUNDARY

Do not infer health, disability, gender or personal activity; do not disclose assistance calls or sensitive feedback; do not promise availability, dignity or suitability for every user; do not certify accessibility, sanitary or welfare compliance; do not authorise facility designation or closure; do not control access or alarms.

WHY THIS IS USEFUL

Correct location, layout, equipment and backup provision before the facility becomes difficult to alter.

Accessibility, inclusion and universal design

AD-035

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which visual, audible and tactile information must be provided so first-time users can find, perceive and understand each critical decision point?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Illuminance and contrast survey, induction-loop field-strength tests, sound-pressure tests using standard signals and sign-recognition trials. Location and coverage: At key route decisions, signs, reception points, alarms and assisted-listening zones, using task-height measurement positions. Observation design: Day and night conditions, occupied ambient-noise periods and controlled testing. Location-level visibility, audibility and task-success summaries. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No gaze tracking, speech recording or disability profiling; laboratory measures do not guarantee comprehension; participant coverage and ambient conditions matter.

AUTHORITATIVE SUPPORTING DATA

Owners: Access consultant, wayfinding owner, fire-alarm owner, equality lead and user representatives. Controlled evidence: Sign strategy, alarm cause-and-effect, language policy, product data, access assessment and consultation results. Precedence: Competent alarm and inclusive-design advice plus user testing over generic contrast or loudness thresholds. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the relevant system owner and responsible designer.

ANALYSIS REQUIRED

Operation: Observation + comparison. Decision task: Multimodal information coverage and comprehension testing. Decision interval: At wayfinding and alarm design, mock-up and commissioning. Minimum evidence: Message set, location, sensory channels, ambient conditions, diverse task participants and acceptance criteria. Method: Map channel redundancy; run scenario-based comprehension tests and identify missed information. Uncertainty: Show participant gaps, learning effects and untested emergency conditions. Safe failure: Do not claim comprehension; redesign or add redundant channels.

ANSWER BOUNDARY

Do not infer sensory or cognitive ability from test performance; do not disclose individual test results or protected information; do not promise universal comprehension; do not certify alarm, accessibility or fire compliance; do not authorise alarm messaging or sign approval; do not control alarms, displays or public-address systems.

WHY THIS IS USEFUL

Simplify messages, improve contrast and lighting, and provide redundant visual, audible and tactile cues.

AD-036

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Where could glare, noise, crowding, flicker, temperature or visual complexity create avoidable sensory load, and what calmer alternative is available?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Calibrated lux, luminance and flicker instruments, non-recording sound-level meters, temperature, humidity and CO2 loggers, and overhead anonymous zone counts. Location and coverage: At consultation-identified routes and rooms, with sensors at seated or standing head height and glare or flicker viewpoints. Observation design: Short-interval logging across daylight, busy and quiet periods in different seasons. 15-minute distributions and exceedance durations by zone. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No audio, video retention, identity or health inference; thresholds vary by person and task; telemetry cannot define sensory comfort without consultation.

AUTHORITATIVE SUPPORTING DATA

Owners: Access and lighting or acoustic consultants, equality lead, estates operators and affected users. Controlled evidence: Inclusive-design brief, environmental criteria, equipment specifications, complaints and consultation outcomes. Precedence: Direct inclusive consultation and competent discipline advice over generic sensor thresholds. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the responsible designer and relevant specialist.

ANALYSIS REQUIRED

Operation: Observation + comparison. Decision task: Sensory-load hotspot mapping and option comparison. Decision interval: During material, lighting, acoustic and circulation design and after occupation. Minimum evidence: Mapped measurements, operating mode, environmental context, proposed finishes and diverse user feedback. Method: Identify coincident stressors and compare zoning, control and refuge options. Uncertainty: Show individual variability, measurement tolerance and unrepresented groups. Safe failure: Describe evidence only and request inclusive testing rather than declaring comfort.

ANSWER BOUNDARY

Do not infer neurodivergence, health condition or distress from behaviour; do not disclose health-related feedback or individual responses; do not promise a sensory-safe environment for everyone; do not certify accessibility, health or comfort compliance; do not authorise scope, access, procurement, construction or operational changes; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Reduce avoidable sensory load, provide calmer routes and add user-controlled or low-stimulus alternatives.

Accessibility, inclusion and universal design

AD-037

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Where are resting places and seating choices needed so route distances, waiting and recovery demands remain manageable?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Surveyed seat inventory and route distances, structured interval seat-use counts or overhead anonymous zone counts, dwell bands and short-term temperature or CO2 logging. Location and coverage: At mapped route intervals, level changes and waiting points; occupancy observations cover seats, not individuals. Observation design: Representative busy and quiet periods, with seasonal checks for external routes. Zone-level occupancy and unavailable-seat durations. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No identity, health, age or purpose inference; use does not reveal unmet need; avoidance may reflect poor comfort or social factors.

AUTHORITATIVE SUPPORTING DATA

Owners: Access consultant, equality lead, estates furniture owner and user representatives. Controlled evidence: Access strategy, seating schedule, route drawings, cleaning requirements and consultation evidence. Precedence: Direct consultation and competent access advice over observed seat use. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the responsible designer and access lead.

ANALYSIS REQUIRED

Operation: Lookup + observation + calculation. Decision task: Rest-point spacing, choice and environmental-suitability assessment. Decision interval: During circulation and furniture design and after layout change. Minimum evidence: Route lengths, gradients, seating locations, seat types, arm and back support, clear spaces and user feedback. Method: Map maximum distances and choice at likely wait points; test alternatives during outages or events. Uncertainty: Show untested needs and seasonal comfort uncertainty. Safe failure: State provision not demonstrated and seek inclusive assessment.

ANSWER BOUNDARY

Do not infer mobility, fatigue, age or medical need from seating use; do not disclose individual access feedback; do not promise that every person can rest when needed; do not certify accessibility or ergonomic adequacy; do not authorise scope, access, procurement, construction or operational changes; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Add well-located, varied seating without obstructing routes or relying on a single distant rest point.

AD-038

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which controls, sockets, handles, switches, screens and service points need different height, force, feedback or approach space for inclusive use?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Measured reach-height survey, activation-force tests, button-use counts and fault-state telemetry. Location and coverage: At each representative control, socket, handle, screen and service point, including clear approach space. Observation design: Prototype or mock-up stage, commissioning and targeted post-occupancy assessment. Control-level dimension, force, failure and assistance summaries. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No identification or inference of ability from repeated attempts; activation counts do not reveal usability; furniture, glare and software interfaces change the task.

AUTHORITATIVE SUPPORTING DATA

Owners: Access consultant, system owner, product supplier, responsible designer and user representatives. Controlled evidence: Control schedule, mounting details, product data, access assessment, mock-up results and interface requirements. Precedence: Competent inclusive-design assessment and direct testing over nominal catalogue dimensions. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the responsible designer and system owner.

ANALYSIS REQUIRED

Operation: Observation + comparison. Decision task: Reach, force, visibility and interaction-sequence evaluation. Decision interval: Before specification, at mock-up and commissioning. Minimum evidence: Installed or prototype geometry, operating force, feedback, approach space, alternative method and diverse user tests. Method: Evaluate complete tasks rather than isolated height; compare manual, automatic and assisted options. Uncertainty: Show participant coverage, setup variance and software dependencies. Safe failure: Do not claim usability; modify or provide an equivalent alternative.

ANSWER BOUNDARY

Do not infer strength, dexterity, disability or competence; do not disclose individual task performance; do not promise universal usability; do not certify accessibility, electrical or product compliance; do not authorise control placement or system setting; do not control connected devices or building systems.

WHY THIS IS USEFUL

Change mounting, interface type, feedback, approach clearance or alternative controls before installation.

Accessibility, inclusion and universal design

AD-039

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Does inclusive access remain equitable during out-of-hours use, examinations, events, maintenance, lift outages and partial closures?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Magnetic door contacts, owner-supplied lift availability, operator-entered temporary-route status, assistance-call response timestamps and overhead anonymous demand counts. Location and coverage: At entrances, temporary routes, doors, lifts, lighting zones and assistance points used in each authorised mode. Observation design: Continuous availability monitoring with scenario-specific assessments. Mode-level availability, detour, wait and assistance summaries. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No identity, credential or disability linkage; availability telemetry does not prove staff response quality or independent usability.

AUTHORITATIVE SUPPORTING DATA

Owners: Security, estates operations, events, timetabling, access lead and responsible person. Controlled evidence: Approved access rules, staffing plans, closure notices, event plans, examination arrangements and reasonable-adjustment processes. Precedence: Current authorised arrangements and direct consultation over assumed equivalence. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the relevant mode owner and equality lead.

ANALYSIS REQUIRED

Operation: Comparison + forecast. Decision task: Equity-by-mode and service-dependency comparison. Decision interval: Before each recurring special mode and after operational policy change. Minimum evidence: Mode definition, affected groups, route and facility availability, staffing or assistance dependency, and alternatives. Method: Compare standard and special modes; identify reduced access, extra disclosure or delay. Uncertainty: Show unknown individual needs and untested rare scenarios. Safe failure: State inequity risk and require owner action rather than suggesting users adapt.

ANSWER BOUNDARY

Do not infer who needs an adjustment or why; do not disclose reasonable-adjustment, credential or attendance information; do not promise equivalent access in every condition; do not certify equality-law or accessibility compliance; do not authorise access rules, closures or adjustments; do not control doors, lifts or alarms.

WHY THIS IS USEFUL

Change operating assumptions, routes, staffing dependencies or physical provision before exclusion becomes routine.

AD-040

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which barriers reported by affected users remain open, who owns each response, and what evidence is needed before an issue can be closed?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: No sensor identifies an accessibility barrier; use consultation and access-assessment evidence, with door, lift or environmental telemetry only to corroborate a reported condition. Location and coverage: At locations raised through consultation or access assessment; telemetry only corroborates a reported condition. Observation design: Continuously available reporting with quarterly and design-gate synthesis. Theme, location, status and recurrence counts with small-group suppression. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: Consent, anonymity and minimum-group thresholds; no disability inference or publication of sensitive narratives; unreported barriers and excluded voices remain possible, and telemetry cannot settle contested needs.

AUTHORITATIVE SUPPORTING DATA

Owners: Equality lead, access consultant, client engagement owner and affected user representatives. Controlled evidence: Consented feedback, access assessment, action register, design responses and decision log. Precedence: Direct user evidence and competent access advice over inferred barriers; signed decisions over informal closure. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the equality lead and responsible designer.

ANALYSIS REQUIRED

Operation: Lookup + comparison + recommendation. Decision task: Barrier traceability, recurrence and closure-quality assessment. Decision interval: At every design stage and scheduled post-occupancy assessment. Minimum evidence: Barrier description, affected task, evidence source, owner, action, due date and user validation where consented. Method: Trace each issue to design response; distinguish resolved, mitigated, accepted and unresolved. Uncertainty: Show representation gaps and contested closure. Safe failure: Keep the item open when user validation or competent evidence is missing.

ANSWER BOUNDARY

Do not infer disability, severity or consent from issue categories; do not disclose individual narratives or protected information; do not promise resolution or universal acceptance; do not certify accessibility or discrimination-law compliance; do not authorise issue closure, adjustment or design change; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Prioritise unresolved barriers, record the decision response and avoid closing an issue merely because a drawing changed.

Daylight, lighting, acoustics, air and comfort

AD-041

ARCHITECTS AND BUILDING DESIGNERS QUESTION

How much useful daylight reaches each occupied workplane through the day and year, and where is electric lighting still required?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Calibrated horizontal and vertical illuminance loggers, external irradiance and blind-position status. Location and coverage: On a defined workplane grid in representative perimeter and deep-plan zones, paired with an unobstructed external irradiance point. Observation design: 5-minute sampling over representative clear, overcast and seasonal periods. Hourly distributions, daylight-availability bands and sensor-quality flags. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No imaging of occupants or activity; furniture, dirt, blind use, sensor cosine error and short seasonal campaigns affect results.

AUTHORITATIVE SUPPORTING DATA

Owners: Lighting designer, architect, estates lighting owner and client brief owner. Controlled evidence: Verified geometry, façade specification, lighting criteria, weather source, luminaire design and approved use. Precedence: Competent lighting calculation and current approved geometry over spot telemetry alone. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the lighting designer and responsible designer.

ANALYSIS REQUIRED

Operation: Calculation + comparison. Decision task: Daylight availability calibration and option comparison. Decision interval: At façade and layout option assessment, detailed design and seasonal validation. Minimum evidence: Verified model, surface reflectance assumptions, local weather file, calibrated measurements or validated simulation. Method: Compare spatial and temporal daylight distributions with task requirements and electric-light dependency. Uncertainty: Show model-measurement error, weather variability and blind-behaviour assumptions. Safe failure: State evidence insufficient where seasons or critical zones are unrepresented.

ANSWER BOUNDARY

Do not infer visual comfort, productivity or wellbeing from illuminance alone; do not disclose restricted drawings, security details or person-level data; do not promise daylight autonomy or energy saving; do not certify lighting, accessibility or health compliance; do not authorise scope, access, procurement, construction or operational changes; do not control blinds or lighting.

WHY THIS IS USEFUL

Adjust window geometry, room depth, surfaces, zoning and lighting controls using seasonal evidence.

AD-042

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which windows, orientations, surfaces or task positions create direct sun, reflected glare or excessive contrast, and which control strategy works without blocking useful daylight?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Vertical illuminance, luminance mapping and blind or shading position, with HDR survey imagery captured only in empty or privacy-masked conditions. Location and coverage: At seated and standing eye positions facing screens, teaching surfaces and circulation views; image capture avoids people. Observation design: Critical solar times across seasons plus representative occupied blind-use periods. Glare-risk intervals and viewpoint summaries. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No identifiable imagery, gaze tracking or screen-content capture; viewpoint choice, sky condition, reflections and user adaptation create large variability.

AUTHORITATIVE SUPPORTING DATA

Owners: Lighting or daylight consultant, architect, façade designer and activity owner. Controlled evidence: Verified geometry, glazing data, shading design, task locations, approved use and consultation evidence. Precedence: Competent glare assessment and direct user testing over generic lux thresholds. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the responsible designer and lighting specialist.

ANALYSIS REQUIRED

Operation: Calculation + estimate. Decision task: Solar geometry, luminance-contrast and shading-option analysis. Decision interval: During façade and internal-layout decisions and post-occupancy tuning. Minimum evidence: Critical viewpoints, task orientation, glazing properties, shading state, weather and representative measurements or simulation. Method: Identify timing and severity of glare; compare external, internal and layout controls. Uncertainty: Show weather, user-position and shading-behaviour sensitivity. Safe failure: Avoid declaring comfort; require further seasonal test where risk remains.

ANSWER BOUNDARY

Do not infer visual ability or discomfort from blind use; do not disclose images containing people, screens or sensitive activity; do not promise glare-free conditions; do not certify visual comfort or lighting compliance; do not authorise façade or control changes; do not control blinds or lighting.

WHY THIS IS USEFUL

Change glazing, external shading, orientation, finishes or task positions before relying on blinds as the only remedy.

Daylight, lighting, acoustics, air and comfort

AD-043

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which luminaire layout, distribution, colour quality, flicker performance and control zoning best supports each task and user group?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Illuminance, spectral or colour-quality spot tests, flicker measurement, circuit power, dimming command and occupancy-zone status. Location and coverage: On task-plane and vertical-plane grids, at representative luminaires and control zones after installation. Observation design: Mock-up, commissioning and selected day or night operating periods. Zone-level light, power, control-response and exception summaries. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No identity or inference of individual preference; meter placement, daylight contribution, driver ageing and control overrides affect results.

AUTHORITATIVE SUPPORTING DATA

Owners: Lighting designer, electrical engineer, client activity owner and estates system owner. Controlled evidence: Lighting calculations, luminaire and driver data, control narrative, room criteria, commissioning results and maintenance plan. Precedence: Competent design and calibrated acceptance tests over unverified product claims or dashboard values. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the lighting designer and system owner.

ANALYSIS REQUIRED

Operation: Observation + comparison. Decision task: Lighting-performance verification and control-zone usability comparison. Decision interval: At mock-up, detailed design, commissioning and seasonal tuning. Minimum evidence: Defined task, target criteria, calibrated measurements, control state, daylight condition and user testing. Method: Compare uniformity, vertical and horizontal light, flicker and control response across scenarios. Uncertainty: Show instrument tolerance, ageing and occupied-condition gaps. Safe failure: Fail the criterion and request redesign or recommissioning rather than averaging away local deficits.

ANSWER BOUNDARY

Do not infer health, visual ability or satisfaction from control use; do not disclose restricted drawings, security details or person-level data; do not promise visual comfort, learning outcomes or energy saving; do not certify lighting or electrical compliance; do not authorise commissioning acceptance or setpoint change; do not control lighting circuits or scenes.

WHY THIS IS USEFUL

Revise luminaire layout, control zoning, interfaces and maintenance assumptions before final acceptance.

AD-044

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which adjacency, construction and junction options provide the required airborne and structure-borne sound separation, including confidentiality?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Non-recording sound-level and vibration meters plus controlled test-source measurements. Location and coverage: On source and receiving sides of named partitions, floors, doors and service penetrations under an acoustician's test plan. Observation design: Unoccupied standardised tests plus occupied ambient campaigns. Frequency-band transmission, vibration and exceedance summaries. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No speech recording, content capture or speaker identification; ambient source variability, doors, flanking paths and incomplete test coverage limit inference.

AUTHORITATIVE SUPPORTING DATA

Owners: Acoustic consultant, architect, MEP engineer and activity owners. Controlled evidence: Acoustic brief, partition and door details, plant data, test certificates, approved uses and complaints history. Precedence: Competent acoustic assessment and standard tests over ambient levels alone. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the acoustic consultant and responsible designer.

ANALYSIS REQUIRED

Operation: Observation + calculation + comparison + bounded diagnosis. Decision task: Source-path-receiver hypothesis testing and separation-option comparison. Decision interval: During adjacency, partition and services coordination and at completion testing. Minimum evidence: Defined source spectra, receiver criteria, construction details, flanking paths and calibrated tests or validated model. Method: Evaluate direct, flanking and structure-borne paths; test door and service states. Diagnosis means ranked, testable hypotheses rather than causal proof. Uncertainty: Show construction variability, source uncertainty and untested paths. Safe failure: State separation not demonstrated and require specialist testing or redesign.

ANSWER BOUNDARY

Do not infer conversation content, research sensitivity or responsibility for noise; do not disclose private speech or sensitive activity; do not promise confidentiality, silence or absence of complaints; do not certify acoustic privacy or compliance; do not authorise scope, access, procurement, construction or operational changes; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Alter adjacencies, constructions, lobbies and service interfaces before defects are concealed.

Daylight, lighting, acoustics, air and comfort

AD-045

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which room volume, absorption, background noise and electroacoustic arrangement best supports speech, collaboration, concentration and assisted listening?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Reverberation-time and speech-transmission tests using standard signals, non-recording ambient sound levels and induction-loop field tests. Location and coverage: At representative listener positions and background-noise locations, plus approved induction-loop test points. Observation design: Mock-up or completed-room tests under empty and representative furnished conditions. Frequency-band RT, STI or equivalent task metrics and background percentiles. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No live speech recording or individual listener testing without consent; furnishing, occupancy, room setup and system settings materially change results.

AUTHORITATIVE SUPPORTING DATA

Owners: Acoustic consultant, AV owner, access consultant and activity owner. Controlled evidence: Acoustic criteria, room geometry, finish schedule, AV design, assistive-listening specification and commissioning plan. Precedence: Competent acoustic and access advice plus standardised tests over subjective impression alone. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the acoustic or AV specialist and responsible designer.

ANALYSIS REQUIRED

Operation: Observation + calculation + comparison. Decision task: Room-acoustic performance prediction and verification. Decision interval: At finish selection, AV coordination, mock-up and commissioning. Minimum evidence: Room volume, absorption, source and receiver positions, background noise, system state and target tasks. Method: Model and measure frequency-dependent performance; compare layouts and occupancy states. Uncertainty: Show model tolerance, furnishing assumptions and listener diversity. Safe failure: Do not accept the space where representative tasks are untested.

ANSWER BOUNDARY

Do not infer hearing ability or comprehension from room metrics; do not disclose individual listening-test data; do not promise speech understanding for every user; do not certify acoustic, AV or accessibility compliance; do not authorise scope, access, procurement, construction or operational changes; do not control AV or assistive-listening systems.

WHY THIS IS USEFUL

Tune volume, absorption, geometry, AV and assistive-listening provision to the intended tasks.

AD-046

ARCHITECTS AND BUILDING DESIGNERS QUESTION

At which facades and opening conditions does external noise conflict with internal use or natural ventilation?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Environmental sound-level meters without audio recording, weather station data and window or vent position. Location and coverage: At selected facade receptors and naturally ventilated openings, with microphone height and weather exposure documented. Observation design: Continuous or campaign monitoring covering weekday, night, weekend and event conditions. Hourly and event-based level distributions with weather exclusions. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No recording of conversations, traffic identifiers or neighbouring activity; short campaigns, unusual construction, wind and unidentified sources limit representativeness.

AUTHORITATIVE SUPPORTING DATA

Owners: Acoustic consultant, planning evidence owner, estates and local environmental records where applicable. Controlled evidence: Site noise survey, planning conditions, façade design, approved operating windows and internal criteria. Precedence: Competent acoustic assessment and formal planning requirements over live dashboard levels. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the acoustic and planning leads.

ANALYSIS REQUIRED

Operation: Observation + comparison. Decision task: External-noise exposure and ventilation-option trade-off analysis. Decision interval: During site planning, façade design and operational-window definition. Minimum evidence: Calibrated survey, weather, façade transmission, opening state, internal task and planning constraints. Method: Model indoor levels for closed, open and mechanically ventilated scenarios; identify critical periods. Uncertainty: Show source, weather and future-development uncertainty. Safe failure: State that natural ventilation viability is unresolved and require specialist assessment.

ANSWER BOUNDARY

Do not infer source identity, neighbour behaviour or nuisance liability; do not disclose private neighbouring information; do not promise quiet conditions or planning acceptance; do not certify noise, planning or façade compliance; do not authorise operating hours or window policy; do not control windows or ventilation.

WHY THIS IS USEFUL

Choose orientation, buffer spaces, façade performance and ventilation strategy with explicit acoustic consequences.

Daylight, lighting, acoustics, air and comfort

AD-047

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which ventilation strategy and air-distribution arrangement maintains acceptable indoor-air evidence across realistic occupancy and operating modes?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Calibrated CO2, temperature, humidity, particulate and selected VOC indicators plus airflow or damper status where commissioned. Location and coverage: In representative occupied breathing zones, at supply or extract terminals and at room thresholds used for occupancy counts. Observation design: 1-minute or 5-minute sampling over representative seasons and modes. 15-minute distributions, event durations and sensor-health flags. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No identity, health diagnosis or activity-content inference; CO2 is a proxy, low-cost pollutant sensors drift, and local sources or windows complicate interpretation.

AUTHORITATIVE SUPPORTING DATA

Owners: MEP engineer, BMS or ventilation owner, estates health-and-safety lead and activity owner. Controlled evidence: Ventilation design, commissioning results, operating schedules, approved occupancy, maintenance records and indoor-air criteria. Precedence: Competent ventilation assessment and current approved use over isolated sensor thresholds. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the MEP engineer and system owner.

ANALYSIS REQUIRED

Operation: Observation + calculation + comparison + bounded diagnosis. Decision task: Ventilation-performance screening and mode comparison. Decision interval: At design, commissioning, seasonal assessment and after occupancy or control change. Minimum evidence: Mapped calibrated sensors, occupancy estimate, outdoor conditions, system state and at least two representative weeks per critical season where feasible. Method: Compare temporal and spatial patterns with design intent; test source, ventilation and control hypotheses. Diagnosis means ranked, testable hypotheses rather than causal proof. Uncertainty: Show proxy limits, sensor error and unmeasured pollutants. Safe failure: Do not declare health or compliance; request competent investigation when evidence is inconsistent.

ANSWER BOUNDARY

Do not infer health, infection, attendance or individual behaviour; do not disclose person-level occupancy or sensitive activity; do not promise healthy air or absence of transmission; do not certify ventilation, workplace or health compliance; do not authorise occupancy or setpoint changes; do not control HVAC or windows.

WHY THIS IS USEFUL

Revise air distribution, zoning, purge strategy, controls and monitoring requirements around actual modes.

AD-048

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Where are overheating, underheating and draught risks most credible, and which passive, system and user-control measures address them?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Air and globe temperature, humidity, air speed, surface temperature, external weather, occupancy bands and control-state telemetry. Location and coverage: At occupied-zone heights near façade, core and known draught paths, paired with local weather and control-state points. Observation design: 1-minute to 5-minute sampling over heating, shoulder and warm periods. 15-minute comfort variables, exceedance durations and control-response summaries. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No health, identity or productivity inference; sensor placement, radiant asymmetry, clothing, activity and adaptive expectations create uncertainty.

AUTHORITATIVE SUPPORTING DATA

Owners: MEP engineer, architect, controls owner, client brief owner and user representatives. Controlled evidence: Thermal model, façade and HVAC design, control narrative, approved use, commissioning data and complaints. Precedence: Competent thermal design and direct feedback over simple temperature thresholds. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the MEP engineer and responsible designer.

ANALYSIS REQUIRED

Operation: Observation + estimate + comparison. Decision task: Thermal-risk, local-discomfort and control-usability analysis. Decision interval: During façade and HVAC option assessment, commissioning and seasonal tuning. Minimum evidence: Validated model or representative measurements, weather, occupancy, system state, local geometry and user-control tests. Method: Compare operative temperature, air movement, solar gains and control response across scenarios. Uncertainty: Show model error, extreme-weather and user-diversity ranges. Safe failure: State risk unresolved and retain mitigation or further seasonal testing.

ANSWER BOUNDARY

Do not infer health, clothing, activity or satisfaction from temperature; do not disclose individual complaints or occupancy; do not promise comfort for every person or weather; do not certify thermal comfort, health or energy compliance; do not authorise setpoint or occupancy changes; do not control HVAC, windows or blinds.

WHY THIS IS USEFUL

Refine façade, zoning, diffusers, shading and user controls before operational fixes become permanent.

Daylight, lighting, acoustics, air and comfort

AD-049

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which junctions and operating conditions create condensation, mould or material-deterioration risk, and how should the detail change?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Surface temperature, relative humidity, dew-point, moisture probes, leak sensors and local weather. Location and coverage: At cold bridges, internal surfaces, concealed-risk interfaces and drainage paths, with dry reference locations. Observation design: Short-interval monitoring through representative cold and wet conditions plus event logging. Daily dew-point margins, wetting and drying cycles, and location-specific alerts. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No occupant health or behaviour inference; surface sensors miss concealed layers; salts, leaks, ventilation and construction moisture can confound patterns.

AUTHORITATIVE SUPPORTING DATA

Owners: Building physicist, architect, MEP engineer, building surveyor and estates defect owner. Controlled evidence: Construction build-ups, hygrothermal analysis, drainage design, condition surveys, maintenance and leak history. Precedence: Competent diagnosis and verified build-up over a single sensor alarm. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the building physicist or surveyor and responsible designer.

ANALYSIS REQUIRED

Operation: Observation + bounded diagnosis. Decision task: Hygrothermal risk and moisture-path evaluation. Decision interval: During envelope detailing, wet-service coordination and after remediation. Minimum evidence: Verified build-up, boundary conditions, moisture source hypotheses, calibrated measurements and seasonal context. Method: Compare dew-point, vapour, thermal bridge and drainage scenarios; test drying potential. Diagnosis means ranked, testable hypotheses rather than causal proof. Uncertainty: Show material-property, concealed-condition and weather uncertainty. Safe failure: Do not diagnose mould or declare safe; request targeted investigation.

ANSWER BOUNDARY

Do not infer mould exposure, health impact or occupant fault; do not disclose health complaints; do not promise condensation-free operation; do not certify fabric, health or waterproofing adequacy; do not authorise remediation or closure; do not control ventilation or heating.

WHY THIS IS USEFUL

Improve thermal continuity, vapour control, drainage, access and monitoring before damage recurs.

AD-050

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which coordinated facade, lighting, acoustic and ventilation option best meets the signed priorities, and what residual penalties must the client accept?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Common-position daylight, electric-light, sound, air and thermal measurements plus calibrated models for each option. Location and coverage: At matched representative zones for each option baseline, using common positions across light, sound, air and thermal domains. Observation design: Matched periods across key seasons and operating modes. Multi-domain 15-minute profiles and criterion-specific exceedance durations. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: Non-image and non-audio evidence by default, with no health or productivity inference; no option is intrinsically best; results depend on agreed criteria, weights, use patterns and model assumptions.

AUTHORITATIVE SUPPORTING DATA

Owners: Architect, lighting, acoustic, façade and MEP leads, client sponsor and user representatives. Controlled evidence: Discipline criteria, option assumptions, design models, cost and carbon information, consultation and risk register. Precedence: Statutory and competent discipline constraints over composite scores; direct consultation over assumed preferences. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the responsible designer and client sponsor.

ANALYSIS REQUIRED

Operation: Comparison + recommendation. Decision task: Multi-objective option comparison and sensitivity analysis. Decision interval: At major option decisions and before performance requirements are frozen. Minimum evidence: At least two viable options, domain-specific evidence, constraints, user priorities and stated weights. Method: Construct a Pareto comparison rather than a single opaque score; test weight and climate sensitivity. Uncertainty: Show non-comparable impacts, trade-offs and evidence gaps. Safe failure: Withhold ranking when constraints are breached or evidence is materially asymmetric.

ANSWER BOUNDARY

Do not infer a universal comfort preference or health outcome; do not disclose restricted drawings, security details or person-level data; do not promise an optimal or complaint-free design; do not certify multidisciplinary compliance or performance; do not authorise option selection or budget; do not control environmental systems.

WHY THIS IS USEFUL

Choose an explicit cross-domain trade-off, record the residual penalty in each losing domain and carry the required mitigation into the specification.

Structure, services, interfaces and maintainability

AD-051

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which proposed loads, openings, supports, fixings and facade interventions are structurally feasible, and which remain conditional on engineer-led verification?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: No autonomous load inference; use engineer-directed measured survey, non-destructive testing and temporary strain or deflection monitoring only under an approved structural investigation. Location and coverage: Only at engineer-approved supports, slabs, beams, columns, fixings and facade anchors affected by the proposal. Observation design: Under an approved test plan before design commitment or during monitored enabling work. Test-case results with calibration, boundary conditions and engineer status. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No unrelated scanning or uncontrolled access to structural-security information; indirect tests, hidden alterations and incomplete load histories cannot establish capacity without engineering judgement.

AUTHORITATIVE SUPPORTING DATA

Owners: Appointed structural engineer and estates structural-record owner. Controlled evidence: Verified drawings, calculations, surveys, material tests, change records and temporary-works information. Precedence: Current structural engineer judgement over telemetry, generic load tables or legacy drawings. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the structural engineer.

ANALYSIS REQUIRED

Operation: Lookup + observation + calculation. Decision task: Structural evidence sufficiency and option screening. Decision interval: Before fixing load locations, penetrations or support strategy. Minimum evidence: Defined load case, location, existing structure, material properties, survey and engineer assessment. Method: Compare option demands with engineer-confirmed constraints; track unknowns and investigation needs. Uncertainty: Show load, material and continuity uncertainty. Safe failure: Do not calculate or state capacity; stop dependent design until competent assessment.

ANSWER BOUNDARY

Do not infer capacity, safety or defect from vibration alone; do not disclose security-sensitive structural details; do not promise feasibility or reserve capacity; do not certify structural adequacy; do not authorise loading, cutting, drilling or temporary works; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Relocate loads, create support zones or commission tests before the layout depends on an unverified assumption.

AD-052

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which electrical, heating, cooling, ventilation, water and data capacities and connection routes are adequate for the proposal, and where is an upgrade decision needed?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Three-phase power-quality loggers or submeters, clamp-on ultrasonic flow instruments, pressure and temperature loggers, valve or damper positions, read-only BMS points and ICT-owner capacity records. Location and coverage: At authorised boards, plant headers, terminals and network or service interfaces that serve the proposed load. Observation design: Representative peak, low and transition periods across relevant seasons. Half-hourly or 15-minute demand percentiles and headroom scenarios. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No network-content inspection, user-level consumption or unauthorised controls; shared loads, diversity, sensor scaling, future projects and abnormal operation can distort headroom.

AUTHORITATIVE SUPPORTING DATA

Owners: MEP engineers, ICT owner, estates energy manager and each system custodian. Controlled evidence: Design duties, schematics, plant schedules, point lists, utility limits, commissioning and approved future-load register. Precedence: Competent capacity study and owner records over dashboard extrapolation. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the relevant discipline engineer and system owner.

ANALYSIS REQUIRED

Operation: Observation + calculation + estimate. Decision task: Capacity-headroom, diversity and upgrade-scenario analysis. Decision interval: At feasibility, developed design and before connections are procured. Minimum evidence: Verified mapping, representative history, design duty, diversity basis, resilience requirement and future committed loads. Method: Normalise and compare observed and rated performance; test simultaneous peak and failure scenarios. Uncertainty: Show metering gaps, climate and coincidence uncertainty. Safe failure: State capacity unverified and require formal study rather than assuming spare headroom.

ANSWER BOUNDARY

Do not infer spare capacity or safe connection from trends alone; do not disclose network topology, credentials or security-sensitive plant details; do not promise connection availability or uninterrupted service; do not certify MEP or ICT capacity and compliance; do not authorise connection, isolation or upgrade; do not control plant, networks, breakers or valves.

WHY THIS IS USEFUL

Resize the proposal, reserve plant space or initiate upgrades before late coordination failure.

Structure, services, interfaces and maintainability

AD-053

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Can each major plant item, equipment unit and facade component be installed, commissioned, removed and replaced through a credible route?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: No continuous sensing is needed; use a current measured survey, component envelopes, route mock-up and owner-confirmed door or lift load data. Location and coverage: Along the complete delivery or removal route, including approach, doors, corners, lifts, hatches and final maintenance zone. Observation design: Survey during design and physical route trial before critical procurement where warranted. Minimum-clearance, turning and dependency records. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No worker tracking or unrestricted publication of secure plant routes; future site conditions, temporary works, packaging and handling methods can change the route envelope.

AUTHORITATIVE SUPPORTING DATA

Owners: Estates asset owner, MEP engineer, structural engineer, logistics lead and responsible designer. Controlled evidence: Asset schedules, manufacturer dimensions, replacement strategy, route drawings, lifting plans and access constraints. Precedence: Competent lifting, structural and logistics advice over geometric fit alone. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the asset owner and relevant engineers.

ANALYSIS REQUIRED

Operation: Lookup + calculation + comparison. Decision task: Replacement-route geometry and dependency analysis. Decision interval: Before plant-room and façade layouts are frozen and before procurement. Minimum evidence: Installed and transport dimensions, mass, route geometry, lifting method, removable elements and outage constraints. Method: Perform swept-volume and sequence checks; identify single-use panels, structural supports and temporary works. Uncertainty: Show tolerance, future obstructions and packaging assumptions. Safe failure: Require a coordinated route trial or specialist lifting study where margin is small.

ANSWER BOUNDARY

Do not infer lifting safety or structural suitability from dimensions; do not disclose secure access routes; do not promise future replaceability; do not certify lifting, structural or installation safety; do not authorise delivery, lifting, removal or temporary works; do not control doors or lifts.

WHY THIS IS USEFUL

Preserve doors, panels, clear zones and lifting points that prevent future destructive access.

AD-054

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Where are architecture, structure, services, fire stopping, ceilings, furniture and installation tolerances competing for the same space?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: No operational sensor detects all clashes; use a coordinated model registered to a dated point cloud and targeted site measurements. Location and coverage: Within the coordinated model at every interface zone, checked against dated point-cloud control and site-verification points. Observation design: Model updates at each coordination cycle and rescan after material installation change. Clash records with evidence status and tolerance. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: Restrict security-sensitive service routes and remove incidental people from survey capture; model completeness, tolerances, movement, installation sequence and hidden services leave residual interface risk.

AUTHORITATIVE SUPPORTING DATA

Owners: BIM information manager, architect and each discipline lead. Controlled evidence: Controlled federated model, verified surveys, design responsibility matrix, builder's work schedule and approved changes. Precedence: Signed discipline information and verified survey over automated clash output. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the BIM manager and responsible designers.

ANALYSIS REQUIRED

Operation: Comparison + bounded diagnosis. Decision task: Rule-based coordination, tolerance and responsibility-interface analysis. Decision interval: At each coordination release and before fabrication. Minimum evidence: Current model versions, element ownership, required clearances, tolerances and evidence status. Method: Detect hard, soft, access and sequencing clashes; assign owner and resolution state. Diagnosis means ranked, testable hypotheses rather than causal proof. Uncertainty: Show unverified geometry, model age and rule limitations. Safe failure: Do not close a clash until accountable owners approve the resolved interface.

ANSWER BOUNDARY

Do not infer constructability or compliance from a clash-free model; do not disclose restricted BIM or security-sensitive services; do not promise zero site clashes; do not certify discipline design or fire stopping; do not authorise model release, fabrication or installation; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Resolve spatial competition visibly and protect maintenance and construction allowances, not just nominal geometry.

Structure, services, interfaces and maintainability

AD-055

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which building-system control and data interfaces are genuinely necessary, who owns them, and how should each fail without unsafe or unmaintainable dependency?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Owner-provided read-only point states, command acknowledgements in a test environment, gateway health and time-synchronisation status. Location and coverage: At approved source and destination control points, gateways and test interfaces; live commands remain disabled outside authorised tests. Observation design: Factory and site acceptance tests plus representative operational trend periods. Interface-level availability, latency and exception summaries. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No network scanning, packet-content inspection, credentials, personal data or unauthorised commands; point names, scaling, clocks, overrides and vendor gateways can misrepresent function.

ANALYSIS REQUIRED

Operation: Lookup + observation + comparison. Decision task: Interface traceability, functional test and maintainability assessment. Decision interval: At specification, integration design, testing and handover. Minimum evidence: Approved use case, source and destination point, units, ownership, failure state, retention and security control. Method: Map data and command paths; test bounded functions, alarms, time and degraded modes. Uncertainty: Show vendor dependencies, unsupported points and test-environment differences. Safe failure: Disable or exclude the interface when authority, security or safe failure is unclear.

AUTHORITATIVE SUPPORTING DATA

Owners: Controls engineer, BMS owner, ICT security owner, MEP engineer and data-protection owner. Controlled evidence: Controls narrative, point list, interface control document, cybersecurity requirements, data dictionary and change records. Precedence: System-owner and cybersecurity approval over discovered or inferred interfaces. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the controls and ICT security owners.

ANSWER BOUNDARY

Do not infer network function or permission from reachable endpoints; do not disclose credentials, topology or security configuration; do not promise interoperability, cybersecurity or vendor support; do not certify cybersecurity, controls or data-protection compliance; do not authorise integration or system access; do not control live building systems outside approved tests.

WHY THIS IS USEFUL

Specify ownership, open interfaces, security boundaries and safe degradation before proprietary dependencies are embedded.

AD-056

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Can maintainers safely find, reach, isolate, test, remove and replace every serviceable component without dismantling unrelated work?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Measured access survey, tool and removal-envelope mock-ups, authorised isolation tests and anonymous task timing. Location and coverage: At each maintainable component and its approach, isolation point, tool envelope and removal path; trials use anonymous task timing. Observation design: Mock-up or coordinated-model assessment before construction, then selected commissioning task trials. Asset-level clearance, access-step and obstruction summaries. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No worker identity, productivity ranking or recording of confidential maintenance activity; mock-ups may not reproduce heat, contamination, emergency conditions, ageing or future replacement tools.

ANALYSIS REQUIRED

Operation: Observation + comparison. Decision task: Task-based maintainability and access-zone validation. Decision interval: During detailed coordination, mock-up and handover. Minimum evidence: Defined maintenance task, frequency, tools, replacement part, isolation, access route and responsible owner. Method: Simulate the complete task; test clearances, sequence, visibility, removal path and reinstatement. Uncertainty: Show future-tool, PPE and degraded-condition uncertainty. Safe failure: State task not demonstrated and redesign or require a physical trial.

AUTHORITATIVE SUPPORTING DATA

Owners: Estates maintenance owner, MEP engineer, contractor or manufacturer and health-and-safety lead. Controlled evidence: Maintenance strategy, O&M; requirements, safe-isolation procedure, asset schedule, access zones and task risk assessments. Precedence: Competent maintenance and safety requirements over geometric model checks. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the asset owner and responsible discipline lead.

ANSWER BOUNDARY

Do not infer worker competence or acceptable task time; do not disclose restricted plant or isolation details; do not promise safe or efficient maintenance; do not certify safe system of work or maintainability compliance; do not authorise maintenance, isolation or access; do not control plant or isolations.

WHY THIS IS USEFUL

Move panels, doors, ceilings and equipment before maintenance requires damage or unsafe improvisation.

Structure, services, interfaces and maintainability

AD-057

ARCHITECTS AND BUILDING DESIGNERS QUESTION

How should each penetration and junction preserve fire, acoustic, air, moisture and security performance, and what installation evidence is required?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: No routine telemetry verifies boundary integrity; use direct inspection, tagged firestop or interface IDs, location-controlled photographs or borescope views, and selected airtightness or acoustic tests. Location and coverage: At every sampled penetration and junction through a rated or environmental boundary, linked to exact asset and location IDs. Observation design: Recorded at installation before concealment and verified at completion. Interface-level status, installer, product, date and inspection provenance. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: Restricted images and security details available only by role; photographs may omit hidden preparation; tags can be misplaced; tests sample only selected interfaces.

AUTHORITATIVE SUPPORTING DATA

Owners: Responsible designer, fire engineer, acoustic or façade specialist, contractor and clerk of works. Controlled evidence: Approved details, product systems, installation method, inspection plan, test evidence and change records. Precedence: Competent inspection and approved system detail over tags, photos or generic product certificates. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the relevant boundary specialist and contract administrator.

ANALYSIS REQUIRED

Operation: Observation + comparison + bounded diagnosis. Decision task: Interface assurance traceability and evidence-completeness check. Decision interval: Before concealment, at completion and after later penetrations. Minimum evidence: Unique interface ID, design detail, installed product, substrate, installer, inspection and deviation resolution. Method: Match each penetration to an approved detail and evidence chain; flag substitutions and missing stages. Diagnosis means ranked, testable hypotheses rather than causal proof. Uncertainty: Show inaccessible hidden work and sampling limits. Safe failure: Mark unverified and require competent inspection or opening-up.

ANSWER BOUNDARY

Do not infer integrity, resistance or security from a tag or image; do not disclose restricted security-boundary details; do not promise performance of concealed work; do not certify fire, acoustic, air, moisture or security compliance; do not authorise concealment, substitution or acceptance; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Prevent untraceable interface failures and retain evidence for future alterations.

AD-058

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which water-supply, drainage, containment and isolation details minimise the consequence of leaks, blockages or uncontrolled discharge?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Zonal flow, pressure, leak cables or point sensors, sump levels, valve state and local moisture or temperature. Location and coverage: At vulnerable wet-service joints, plant pans, risers, low points, drains and downstream damage-sensitive zones. Observation design: Continuous event sensing after commissioning plus representative discharge or rainfall tests. Event time, duration, zone and sensor-health summaries. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No inference of individual water use; sensor placement, shared branches, condensation and unobserved slow leaks create false negatives or positives.

AUTHORITATIVE SUPPORTING DATA

Owners: Public-health engineer, estates water and drainage owners, building surveyor and risk owner. Controlled evidence: Schematics, drainage survey, isolation schedule, leak history, asset criticality and maintenance plans. Precedence: Competent hydraulic and fabric assessment over leak alerts or flow anomalies. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the relevant system owner and engineer.

ANALYSIS REQUIRED

Operation: Observation + bounded diagnosis. Decision task: Consequence-weighted leak path, drainage capacity and isolation-zone analysis. Decision interval: During wet-service and roof design, commissioning and after incidents. Minimum evidence: Verified routes, gradients or capacity, isolation boundaries, vulnerable assets, sensor coverage and test results. Method: Map source-path-consequence; test blocked, burst and isolation scenarios; identify detection gaps. Diagnosis means ranked, testable hypotheses rather than causal proof. Uncertainty: Show hidden routes, blockage likelihood and sensor coverage uncertainty. Safe failure: Do not declare safe; require survey, test or redesign where consequence is high.

ANSWER BOUNDARY

Do not infer source, fault or misuse from flow alone; do not disclose sensitive asset or security information; do not promise leak prevention or zero damage; do not certify water, drainage or waterproofing compliance; do not authorise isolation, shutdown or remediation; do not control valves, pumps or water systems.

WHY THIS IS USEFUL

Provide accessible isolation, drainage falls, containment, leak detection and protected routes around critical spaces.

Structure, services, interfaces and maintainability

AD-059

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which commissioning tests, seasonal checks, training and asset information must be specified so the building can be operated as designed?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Calibrated portable test instruments and commissioned permanent points named in the commissioning plan, plus BMS functional-test records, alarms, trends, meters and interface results. Location and coverage: At each commissioned system, representative terminal zone and handover test point named in the commissioning plan. Observation design: Pre-functional checks, witnessed functional tests, occupation transition and seasonal assessment. Criterion-level pass, fail, exception and retest records. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No person-level training analytics or unauthorised live commands; test conditions may not represent peak season, occupancy or future degradation.

AUTHORITATIVE SUPPORTING DATA

Owners: Commissioning manager, contract administrator, system owners, responsible designers and client operations. Controlled evidence: Commissioning plan, design intent, test scripts, training plan, O&M; information, asset data and defects register. Precedence: Witnessed competent tests and contractual requirements over dashboard status. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the commissioning manager and contract administrator.

ANALYSIS REQUIRED

Operation: Lookup + observation + comparison. Decision task: Commissioning coverage, requirement traceability and readiness assessment. Decision interval: Before handover, at seasonal milestones and after material system change. Minimum evidence: Requirement, test method, calibrated instrument, witness, result, deviation, owner and retest status. Method: Trace every requirement to evidence; distinguish incomplete, failed, deferred and seasonally pending items. Uncertainty: Show untested conditions and temporary overrides. Safe failure: Do not mark ready; retain open action and operational mitigation.

ANSWER BOUNDARY

Do not infer operator competence or user acceptance from attendance; do not disclose restricted system or staff information; do not promise handover, acceptance or future performance; do not certify completion, compliance or system safety; do not authorise handover, payment or closure; do not control systems outside approved tests.

WHY THIS IS USEFUL

Procure complete acceptance evidence, organise training and keep seasonal duties visible beyond practical completion.

AD-060

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which phasing and temporary-work strategy keeps routes, services, environmental conditions and life-safety boundaries workable in an occupied building?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Calibrated noise, dust or PM, temperature, CO2 and vibration instruments at work boundaries, plus owner-approved temporary route and service states. Location and coverage: At site or public boundaries, temporary routes, occupied rooms beside works and temporary-service interfaces defined by the construction plan. Observation design: Continuous or task-based monitoring during authorised work windows. Shift and event summaries against project action levels. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No audio, images, identity or worker performance inference; telemetry does not replace the construction phase plan, risk controls, permits, competent supervision or statutory monitoring.

AUTHORITATIVE SUPPORTING DATA

Owners: Principal designer, principal contractor, client operations, security, safety and affected service owners. Controlled evidence: Construction phase plan, phasing drawings, permits, temporary works, isolation plans, decant and communication arrangements. Precedence: Approved construction and safety controls over telemetry; current closure notices over historic plans. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the principal contractor and client dutyholders.

ANALYSIS REQUIRED

Operation: Forecast + comparison + recommendation. Decision task: Phase-interface, temporary-service and occupied-impact scenario assessment. Decision interval: Before each phase and after material sequencing change. Minimum evidence: Work scope, occupied activities, temporary routes, service dependencies, monitoring limits and escalation owners. Method: Map time and space conflicts; test outages, dust or noise controls and recovery sequence. Uncertainty: Show rare-event, contractor-method and occupancy uncertainty. Safe failure: Stop recommendation and escalate where approved safe controls or authority are absent.

ANSWER BOUNDARY

Do not infer worker compliance, competence or occupant tolerance from sensors; do not disclose sensitive activities or security arrangements; do not promise safe, quiet or disruption-free construction; do not certify construction safety, permits or temporary works; do not authorise work, isolation, closure or occupancy; do not control temporary or permanent systems.

WHY THIS IS USEFUL

Refine phasing, decant, temporary partitions and service continuity before work packages are committed.

Materials, lifecycle, adaptability and carbon

AD-061

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which existing fabric, components and finishes should be retained or reused, and what condition or performance evidence is needed before that decision is fixed?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Measured or laser survey, calibrated moisture and thermal checks, borescope inspection, material identification, targeted samples and specialist non-destructive tests where required. Location and coverage: At representative retained elements, interfaces and suspect areas, with sample locations fixed to the condition-survey grid. Observation design: Survey before strip-out and updated after opening-up. Component-level condition, dimensions, test provenance and reuse status. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: Restricted imagery only; no occupant data; hidden fixings, contamination, undocumented treatment and destructive removal can change suitability.

AUTHORITATIVE SUPPORTING DATA

Owners: Architect, building surveyor, structural engineer, client asset owner and specialist material assessor. Controlled evidence: Condition survey, product and maintenance history, hazardous-material register, performance requirements and reuse inventory. Precedence: Competent condition and performance assessment over visual appearance or carbon benefit alone. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the responsible designer and relevant specialist.

ANALYSIS REQUIRED

Operation: Observation + bounded diagnosis. Decision task: Reuse suitability, intervention and residual-risk comparison. Decision interval: At feasibility, strip-out design and after opening-up. Minimum evidence: Component identity, condition, required performance, removal method, warranty implications and alternative. Method: Classify retain as-is, remediation, relocate, downcycle or reject; compare carbon, cost and risk. Diagnosis means ranked, testable hypotheses rather than causal proof. Uncertainty: Show hidden-condition and future-performance uncertainty. Safe failure: Do not allocate to reuse where safety, provenance or performance is unverified.

ANSWER BOUNDARY

Do not infer material safety, strength or compliance from age or appearance; do not disclose restricted survey or hazard data; do not promise reuse yield or future life; do not certify material, structural, fire or health performance; do not authorise removal, reuse or disposal; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Protect valuable existing fabric, target testing and avoid designing around components that cannot credibly perform.

AD-062

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which material options best withstand the expected traffic, cleaning, moisture, impact and maintenance regime without disproportionate replacement?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Overhead thermal or time-of-flight footfall counts, cleaning-task records, humidity, wetting or leak sensors, impact or fault logs, controlled material tests and standardised condition scoring. Location and coverage: At high-traffic, wet, impact-prone and cleaning-intensive zones, plus controlled material test specimens. Observation design: Representative baseline and periodic post-installation inspections. Monthly exposure and defect-rate summaries by material zone. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No worker performance or individual behaviour profiling; past exposure may not match future use; cleaning products and workmanship materially affect durability.

AUTHORITATIVE SUPPORTING DATA

Owners: Architect, estates maintenance and cleaning owners, supplier and client asset manager. Controlled evidence: Material data, warranty, cleaning regime, maintenance history, condition surveys and approved aesthetic requirements. Precedence: Competent specification and owner maintenance requirements over marketing claims or isolated telemetry. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the responsible designer and asset owner.

ANALYSIS REQUIRED

Operation: Observation + comparison + estimate. Decision task: Exposure-class, maintenance-burden and option comparison. Decision interval: During specification and scheduled post-occupancy assessment. Minimum evidence: Expected exposure, cleaning method, maintainability, warranty, representative defect data and service-life assumptions. Method: Compare options on failure modes, access, replacement and appearance tolerance. Uncertainty: Show workmanship, policy and future-use uncertainty. Safe failure: State evidence weak and require sample, mock-up or reference-site inspection.

ANSWER BOUNDARY

Do not infer misuse or staff performance from wear; do not disclose individual incident information; do not promise service life or appearance retention; do not certify product suitability or warranty; do not authorise material approval or procurement; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Select robust, cleanable and maintainable materials with realistic maintenance commitments.

Materials, lifecycle, adaptability and carbon

AD-063

ARCHITECTS AND BUILDING DESIGNERS QUESTION

What is the whole-element embodied-carbon difference between retaining, restoring, reusing and replacing each major component under consistent assumptions?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: No direct sensor establishes embodied carbon; use survey-verified quantities, measured weights, tagged reuse or waste streams and controlled carbon factors. Location and coverage: At surveyed element extents, weighing points and tagged reuse or waste streams; carbon factors remain documentary inputs. Observation design: Quantity take-off at each design gate and measured reconciliation after strip-out or construction. Element-level mass, source, reuse and waste summaries. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No personal data; commercial quantities remain role-restricted where required; generic factors, uncertain quantities, allocation rules, incomplete declarations and future reuse yields can dominate the result.

AUTHORITATIVE SUPPORTING DATA

Owners: Client carbon lead, architect, cost consultant, supplier and quantity-information owner. Controlled evidence: Controlled quantities, EPDs, product data, transport and waste assumptions, reuse inventory and project carbon methodology. Precedence: Project-approved methodology and product-specific verified data over generic averages where available. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the carbon lead and responsible designer.

ANALYSIS REQUIRED

Operation: Calculation + comparison. Decision task: Whole-life carbon option calculation and sensitivity analysis. Decision interval: At concept, developed design, pre-tender and as-built reporting. Minimum evidence: Consistent boundary, quantities, life-cycle modules, data quality, service-life and end-of-life assumptions. Method: Calculate alternatives on a like-for-like functional basis; separate retained, new, biogenic and uncertain contributions. Uncertainty: Show data-quality classes and sensitivity to quantity, factor and life assumptions. Safe failure: Report a range or incomparable options when boundaries or functions differ.

ANSWER BOUNDARY

Do not infer environmental superiority from one module or carbon value; do not disclose commercially restricted supplier data; do not promise net-zero or verified reduction; do not certify EPD, carbon claim or regulatory compliance; do not authorise product selection or procurement; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Direct redesign or retention effort to elements with material carbon benefit and avoid claims supported by weak data.

AD-064

ARCHITECTS AND BUILDING DESIGNERS QUESTION

How does each design option change operational energy and carbon under realistic weather, occupancy, operating hours and control behaviour?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Verified electricity, heat and cooling submeters, plant power or flow points, calibrated room temperature and CO2 loggers, an external weather station, anonymous occupancy bands and read-only control states. Location and coverage: At affected end-use meters, plant assets, representative room zones and an external weather station. Observation design: 5-minute to half-hourly data over representative seasons, with pre-change baseline where possible. Weather- and occupancy-normalised monthly and peak-demand summaries. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No person-level energy attribution; submeter gaps, control changes, atypical weather and model calibration can obscure option effects.

AUTHORITATIVE SUPPORTING DATA

Owners: Energy manager, MEP engineer, controls owner, client carbon lead and utility data owner. Controlled evidence: Validated energy model, meter hierarchy, tariffs or carbon factors, operating schedules, design intent and commissioning data. Precedence: Approved model and verified meter mapping over whole-building averages. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the MEP and carbon leads.

ANALYSIS REQUIRED

Operation: Estimate + forecast + comparison. Decision task: Weather- and use-normalised operational-energy scenario estimate and comparison. Decision interval: At option selection, design freeze and seasonal post-occupancy assessment. Minimum evidence: Consistent boundary, weather, occupancy modes, system efficiency, controls and mapped baseline. Method: Compare end-use demand, peaks and carbon under stated scenarios; calibrate where evidence permits. Uncertainty: Show model error, rebound, future-grid and behavioural sensitivity. Safe failure: Do not attribute savings where baseline or intervention isolation is inadequate.

ANSWER BOUNDARY

Do not infer individual behaviour or responsibility for energy use; do not disclose tenant or person-level consumption; do not promise bill, energy or carbon savings; do not certify energy performance or regulatory compliance; do not authorise operational setpoints or investment; do not control energy systems.

WHY THIS IS USEFUL

Compare fabric, façade, zoning and servicing options without hiding assumptions behind a single annual figure.

Materials, lifecycle, adaptability and carbon

AD-065

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which assemblies can be disassembled, maintained and reused without destructive access, bonded contamination or loss of evidence?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Component tags, connection surveys, pilot-disassembly timing and condition checks. Location and coverage: At representative assemblies, connections and high-value components during mock-up and trial disassembly. Observation design: Prototype or mock-up stage, selected installation checks and end-of-life trials where possible. Component-level connection, tool, time, damage and recovery summaries. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No worker identity or productivity scoring; pilot conditions may not reflect ageing, corrosion, adhesives or future access.

AUTHORITATIVE SUPPORTING DATA

Owners: Architect, contractor, manufacturer, asset owner and circular-economy adviser. Controlled evidence: Detail drawings, product assembly information, material passport, warranty, maintenance and end-of-life instructions. Precedence: Verified installed detail and competent product advice over design intent alone. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the responsible designer and asset owner.

ANALYSIS REQUIRED

Operation: Observation + comparison. Decision task: Design-for-disassembly and recoverability assessment. Decision interval: At detail design, product selection and selected post-installation assessment. Minimum evidence: Connection type, access, sequence, tools, damage risk, material compatibility and future reuse pathway. Method: Score reversibility, separability and replacement independence; test representative assemblies. Uncertainty: Show ageing, market and future-regulation uncertainty. Safe failure: Do not claim recoverability where installed evidence or feasible pathway is absent.

ANSWER BOUNDARY

Do not infer future reuse market or component condition; do not disclose commercially sensitive product details; do not promise reuse, residual value or waste avoidance; do not certify circularity claim or product warranty; do not authorise product approval, dismantling or reuse; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Replace adhesive, composite or concealed details with reversible, legible and maintainable assemblies.

AD-066

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which structural grid, partition, service-zone and connection strategy preserves credible future uses with the least waste and downtime?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Overhead anonymous counts, magnetic contacts on movable partitions, tagged furniture-state surveys, approved service-outlet or controller events, manual changeover timing and controlled work records. Location and coverage: At adaptable-zone thresholds, movable elements, service-connection points and documented change-event locations. Observation design: Quarterly or annual post-occupancy summaries and each approved change event. Configuration frequency, downtime, waste and unresolved-interface summaries. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No team performance or activity-content inference; past churn is not a forecast; undocumented changes and policy shifts alter apparent adaptability.

AUTHORITATIVE SUPPORTING DATA

Owners: Client strategy owner, estates asset manager, architect and MEP or ICT owners. Controlled evidence: Future-use scenarios, grid and servicing strategy, room data sheets, change records, component passports and cost data. Precedence: Approved strategic scenarios and competent discipline constraints over extrapolated use trends. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the client sponsor and responsible designer.

ANALYSIS REQUIRED

Operation: Forecast + comparison. Decision task: Adaptation scenario, conversion effort and performance-retention comparison. Decision interval: At briefing, option selection and major future change. Minimum evidence: At least three future-use scenarios, spatial and service requirements, component reuse, downtime and performance constraints. Method: Simulate conversion steps, retained value, waste and new dependencies. Uncertainty: Show scenario and market uncertainty. Safe failure: Avoid a future-proof claim; retain multiple pathways when no scenario dominates.

ANSWER BOUNDARY

Do not infer future organisational use from current patterns; do not disclose restricted drawings, security details or person-level data; do not promise future-proofing, low-cost change or retained performance; do not certify future compliance or capacity; do not authorise change of use or capital investment; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Choose structural grids, service zones and reversible partitions that preserve credible options.

Materials, lifecycle, adaptability and carbon

AD-067

ARCHITECTS AND BUILDING DESIGNERS QUESTION

How will hotter summers, intense rain, wind and repeated wetting or freeze-thaw exposure affect the selected materials and details over their intended life?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: A local weather station for rain, wind, temperature and humidity, plus surface temperature or moisture probes, drain-level or leak sensors, crack gauges and specialist exposure coupons where justified. Location and coverage: At representative exposed facade, roof, entrance and external-work interfaces, with local weather and wetting reference points. Observation design: Continuous or seasonal monitoring of representative exposures with event capture. Event severity, wetting-drying and exceedance summaries. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No personal data; local sensors are short-term; future extremes and workmanship remain uncertain.

AUTHORITATIVE SUPPORTING DATA

Owners: Client climate-risk owner, architect, façade or drainage specialist and estates maintenance. Controlled evidence: Approved climate scenarios, weather files, flood or drainage studies, product exposure limits, condition and maintenance data. Precedence: Project-approved climate basis and competent specialist design over recent weather alone. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the climate-risk owner and responsible designer.

ANALYSIS REQUIRED

Operation: Forecast + estimate. Decision task: Climate-stress, material-response and adaptation-pathway assessment. Decision interval: At material and detail selection and periodic asset assessment. Minimum evidence: Service life, location, exposure, climate scenario, material limits, maintenance and consequence of failure. Method: Test low, central and high climate cases; identify thresholds and no-regret details. Uncertainty: Show model, scenario and compound-event uncertainty. Safe failure: Do not claim resilience where the scenario basis or specialist evidence is absent.

ANSWER BOUNDARY

Do not infer future exact weather or material life; do not disclose restricted drawings, security details or person-level data; do not promise climate resilience or no damage; do not certify façade, drainage or product compliance; do not authorise climate-risk acceptance or investment; do not control drainage or environmental systems.

WHY THIS IS USEFUL

Change exposure, detailing, drainage, replaceability and maintenance before climate risk is locked in.

AD-068

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which products have decision-grade evidence for responsible sourcing, health, emissions, provenance and intended-use suitability?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: No routine sensor proves product provenance; use batch identifiers and controlled declarations, with specialist post-installation VOC sampling only where specified. Location and coverage: At product delivery or installation records and, where specified, post-installation room sampling points selected by an indoor-air specialist. Observation design: Document check before approval, delivery reconciliation and defined post-installation sampling. Product-level evidence status and room-level screening summaries. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No health diagnosis, worker exposure profiling or publication of commercially restricted supply data; certificates can expire, chain-of-custody can break and room emissions can have multiple sources.

AUTHORITATIVE SUPPORTING DATA

Owners: Client procurement and sustainability owners, architect, supplier and health-and-safety adviser. Controlled evidence: Product declarations, chain-of-custody, responsible-sourcing certificates, safety data, emissions tests and substitution records. Precedence: Current verified product-specific evidence and client policy over marketing claims. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the procurement owner and responsible designer.

ANALYSIS REQUIRED

Operation: Lookup + comparison. Decision task: Product-evidence completeness, applicability and substitution comparison. Decision interval: Before specification approval, procurement and substitution. Minimum evidence: Exact product, manufacturer, batch or certificate scope, intended use, validity period and required criterion. Method: Verify scope, date and provenance; compare alternatives and identify unsupported claims. Uncertainty: Show missing modules, unverified supply-chain stages and mixed-source room emissions. Safe failure: Reject or mark unresolved when evidence is expired, inapplicable or inaccessible.

ANSWER BOUNDARY

Do not infer worker conditions, product healthfulness or absence of risk from a certificate; do not disclose confidential supplier information; do not promise ethical sourcing or healthy indoor air; do not certify product claims, emissions or legal compliance; do not authorise product approval or procurement; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Select products with traceable evidence and prevent unsupported substitutions from entering the project.

Materials, lifecycle, adaptability and carbon

AD-069

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which option provides the best whole-life value after maintenance access, replacement cycles, downtime, energy, carbon and residual risk are included?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: No dedicated new sensor is assumed; use CMMS work orders and faults, verified energy or water meters, anonymous use counts, condition surveys and owner-approved downtime records from comparable assets. Location and coverage: At comparable assets and spaces within the option boundary; financial and carbon records stay in owner systems. Observation design: Multi-year history where available, supplemented by scenario assumptions. Annualised cost, downtime, fault and consumption summaries. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No worker performance or individual-use attribution; historic data may reflect different maintenance standards, tariffs, occupancy and climate.

AUTHORITATIVE SUPPORTING DATA

Owners: Client asset manager, cost consultant, estates maintenance, energy owner and responsible designer. Controlled evidence: Capital estimate, maintenance strategy, service lives, replacement access, utilities, risk allowances and residual-value assumptions. Precedence: Client-approved appraisal method and current cost evidence over vendor headline savings. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the client asset manager and cost consultant.

ANALYSIS REQUIRED

Operation: Calculation + forecast + comparison. Decision task: Whole-life cost and value sensitivity analysis. Decision interval: At option selection, pre-tender and major substitution. Minimum evidence: Consistent study period, discounting, capital, maintenance, replacement, downtime, energy and risk assumptions. Method: Compare present value and non-monetary service impacts; test key assumptions and break-even points. Uncertainty: Show cost range, data age and correlated risks. Safe failure: Do not rank options where functional outcomes differ or cost evidence is not comparable.

ANSWER BOUNDARY

Do not infer future tariffs, failure rates or maintenance behaviour as certain; do not disclose commercially confidential costs; do not promise savings, payback or budget; do not certify investment value or procurement compliance; do not authorise option selection or expenditure; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Choose an option with visible lifecycle consequences rather than the lowest first cost.

AD-070

ARCHITECTS AND BUILDING DESIGNERS QUESTION

What material and component information must be handed over so future teams can locate, inspect, maintain, replace, reuse and recover each item?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: QR or RFID component identifiers, a current laser or measured location survey, delivery and installation records, quantity reconciliation and controlled update events for removed or relocated items. Location and coverage: On selected maintainable, high-value, hazardous or reusable components, with tags linked to surveyed installation locations. Observation design: Captured at delivery and installation, verified at handover and updated on change. Component-level identity, location, material, evidence and status. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: Role-based access for security-sensitive assets; no personal data; tags can detach, records can become stale and product provenance may be incomplete.

AUTHORITATIVE SUPPORTING DATA

Owners: Asset information manager, contractor, suppliers, client asset owner and responsible designer. Controlled evidence: Asset information requirements, product data, EPDs, warranties, installation records, hazardous-material data and change register. Precedence: Verified installed information and owner governance over design-stage assumptions. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the asset information manager and asset owner.

ANALYSIS REQUIRED

Operation: Lookup + observation + comparison. Decision task: Material-passport completeness, provenance and updateability assessment. Decision interval: At information-requirement definition, handover and each material alteration. Minimum evidence: Unique item or assembly, installed location, composition, source, disassembly, hazard, warranty and owner. Method: Validate mandatory fields, source links, conflicts and update responsibility; sample against physical installation. Uncertainty: Show unverified fields and future market uncertainty. Safe failure: Mark unknown rather than populating with generic product data.

ANSWER BOUNDARY

Do not infer future recyclability, value or condition from a passport; do not disclose security-sensitive asset locations or supplier-confidential data; do not promise recovery, resale or circular outcome; do not certify passport accuracy or end-of-life compliance; do not authorise disposal, reuse or data release; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Reduce future destructive survey, preserve provenance and make maintenance, reuse and recovery decisions faster and safer.

Post-occupancy evidence, design validation and lessons

AD-071

ARCHITECTS AND BUILDING DESIGNERS QUESTION

How does observed occupancy compare with the design assumptions for each space, peak, operating mode and shared-resource dependency?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Commissioned overhead thermal or time-of-flight people counters, booking events, magnetic door contacts and CO2 only as a secondary occupancy proxy. Location and coverage: At each evaluated room threshold and shared-resource entrance, with representative CO2 probes away from doors and supply jets. Observation design: At least four representative weeks per critical mode and season where feasible. 15-minute distributions, peak percentiles and mismatch bands. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No identity, attendance, productivity or device tracking; counter error, no-shows, open doors, changed use and exceptional periods reduce comparability.

AUTHORITATIVE SUPPORTING DATA

Owners: Estates space manager, timetabling or booking owner, client sponsor and responsible designer. Controlled evidence: Approved design assumptions, capacity and use records, current timetable, booking policy, closures and change register. Precedence: Current approved use and competent capacity advice over historical labels or telemetry alone. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the space manager and client sponsor.

ANALYSIS REQUIRED

Operation: Observation + comparison. Decision task: Assumption-to-outcome comparison and demand-shift classification. Decision interval: First seasonal post-occupancy assessment and after material use change. Minimum evidence: Original assumption, mapped current use, calibrated data, representative period and known operational changes. Method: Compare distribution not just average; separate design mismatch, management practice and later change. Uncertainty: Show counting and attribution uncertainty. Safe failure: State the assumption cannot be validated where mapping or current use is unknown.

ANSWER BOUNDARY

Do not infer individual attendance, purpose or performance; do not disclose person-level booking or access data; do not promise capacity or future demand; do not certify room occupancy, fire capacity or design success; do not authorise change of use or occupancy; do not control access or booking.

WHY THIS IS USEFUL

Correct future briefs, rebalance space or investigate operating causes without attributing differences to individual users.

AD-072

ARCHITECTS AND BUILDING DESIGNERS QUESTION

How do observed route lengths, waits and hand-offs compare with the workflow and adjacency assumptions used in the design?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Anonymous threshold counts, queue observations and consented task-time trials at the interfaces named in the original workflow rationale. Location and coverage: At the exact hand-offs, thresholds and shared resources named in the original adjacency rationale. Observation design: Representative normal and peak periods after stabilisation. Transition, wait, assistance and detour summaries without linked journeys. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No linked journeys, identity, activity-content capture or productivity inference; changed procedures, staffing and novelty effects can dominate observed route or wait differences; causation is not established.

AUTHORITATIVE SUPPORTING DATA

Owners: Activity owners, estates operations, client sponsor and responsible designer. Controlled evidence: Design rationale, approved workflow, occupancy changes, service records, complaints and consultation. Precedence: Current authorised workflow and direct user evidence over movement frequency alone. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the activity owner and client sponsor.

ANALYSIS REQUIRED

Operation: Observation + comparison. Decision task: Design-intent outcome and contributing-factor analysis. Decision interval: Seasonal post-occupancy assessment and after major workflow change. Minimum evidence: Stated adjacency objective, baseline or comparator, current workflow, mapped observations and stakeholder feedback. Method: Compare expected and observed task paths; test alternative explanations and identify controllable factors. Uncertainty: Show limited baseline and confounding operational changes. Safe failure: Avoid causal claims and record insufficient evidence where no comparator exists.

ANSWER BOUNDARY

Do not infer collaboration, efficiency or staff performance from movement; do not disclose sensitive workflow or complaints; do not promise productivity or satisfaction gains; do not certify design effectiveness or operational adequacy; do not authorise space reallocation or process change; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Test whether the adjacency rationale still fits real workflows and target a layout or operational change without claiming causation.

Post-occupancy evidence, design validation and lessons

AD-073

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which accessibility barriers remain after occupation, including those introduced by furniture, operations, outages, security changes or temporary routes?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: No sensor certifies accessibility; use affected-user consultation, measured route checks and owner logs for doors, lifts and assistance systems. Location and coverage: Along required routes and at doors, lifts, facilities and assistance points, prioritising locations reported by affected users. Observation design: Continuous fault monitoring plus scheduled inclusive walkthroughs and mode-specific assessments. Barrier type, location, recurrence, duration and action status. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No disability inference, identity linkage or broad disclosure of sensitive narratives; telemetry cannot establish dignity, equity or barriers that are unreported or outside the monitored route.

AUTHORITATIVE SUPPORTING DATA

Owners: Equality lead, access consultant, estates operations and affected user representatives. Controlled evidence: Access strategy, commissioning evidence, fault records, furniture changes, closure plans, feedback and action register. Precedence: Direct user evidence and competent access advice over sensor status. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the equality lead and responsible designer.

ANALYSIS REQUIRED

Operation: Lookup + observation + bounded diagnosis. Decision task: Post-occupancy barrier traceability and recurrence analysis. Decision interval: Early occupation, seasonal assessment and after route or policy change. Minimum evidence: Approved access intent, current physical and operational state, user feedback, owner and action evidence. Method: Compare evidence against design, installation, maintenance and operational hypotheses; verify closure with users where consented. Diagnosis means ranked, testable hypotheses rather than causal proof. Uncertainty: Show representation and reporting gaps. Safe failure: Keep unresolved where competent or user validation is missing.

ANSWER BOUNDARY

Do not infer disability, severity or consent; do not disclose individual adjustment or feedback data; do not promise barrier removal or universal access; do not certify accessibility or equality compliance; do not authorise issue closure, route or operational change; do not control doors or lifts.

WHY THIS IS USEFUL

Learn which details and operational dependencies failed and update future designs or remedial priorities.

AD-074

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Do observed daylight, electric lighting, acoustics, air quality and thermal conditions support the intended activities across seasons and operating modes?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Co-located calibrated lux, non-recording sound-level, temperature, humidity, CO2 and selected PM instruments, an external weather station, overhead anonymous occupancy counts and read-only control states. Location and coverage: In representative occupied zones and edge conditions, with co-located instruments and documented mounting heights. Observation design: Short-interval seasonal campaigns after systems stabilise. Criterion-specific distributions, exceedance durations and sensor-health indicators. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No audio, images, identity, health or productivity inference; sensor placement, altered use, overrides, maintenance and unusual weather complicate attribution.

AUTHORITATIVE SUPPORTING DATA

Owners: Responsible discipline designers, estates system owners, commissioning manager and client sponsor. Controlled evidence: Design criteria, models, commissioning results, control narrative, current use, maintenance and consented feedback. Precedence: Competent discipline assessment and current use over dashboard thresholds. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the relevant system owner and responsible designer.

ANALYSIS REQUIRED

Operation: Observation + comparison + bounded diagnosis. Decision task: Seasonal performance comparison and contributing-condition hypothesis testing. Decision interval: At agreed post-occupancy seasons and after material system change. Minimum evidence: Design intent, calibrated data, system and control states, weather, occupancy mode and user feedback. Method: Compare spatial and temporal outcomes with intent; test design, commissioning, operation and use hypotheses. Diagnosis means ranked, testable hypotheses rather than causal proof. Uncertainty: Show sensor, model and attribution uncertainty. Safe failure: Do not declare compliance or cause; request specialist investigation where evidence conflicts.

ANSWER BOUNDARY

Do not infer health, satisfaction or performance from environmental data; do not disclose individual complaints or occupancy; do not promise comfort or healthy conditions; do not certify environmental compliance or design performance; do not authorise setpoint or system changes; do not control lighting, HVAC, blinds or windows.

WHY THIS IS USEFUL

Tune systems, correct defects and update future performance assumptions with seasonal evidence.

Post-occupancy evidence, design validation and lessons

AD-075

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Did the project achieve its intended energy and water outcomes after weather, occupancy, operating hours, controls and later changes are normalised?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Fiscal or class-appropriate energy and water meters, verified end-use submeters, external weather data, anonymous occupancy or booking bands, read-only system states and leak-event records. Location and coverage: At the agreed project boundary, incoming meters, material end-use submeters and relevant plant or control points. Observation design: Half-hourly or finer data for baseline and at least one representative post-occupancy year where available. Monthly normalised demand, peaks, baseload and data-quality summaries. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No person-level consumption or behavioural blame; meter mapping, tariff or carbon changes, later projects and control drift can obscure comparison.

AUTHORITATIVE SUPPORTING DATA

Owners: Energy and water managers, MEP designer, controls owner, client carbon lead and finance or utility data owners. Controlled evidence: Performance targets, meter hierarchy, baseline method, commissioning, operating schedules, changes and verified bills. Precedence: Verified billing and meter mapping plus agreed methodology over dashboard totals. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the energy or water manager and client sponsor.

ANALYSIS REQUIRED

Operation: Calculation + comparison. Decision task: Measurement and verification with bounded attribution. Decision interval: Monthly monitoring, seasonal assessment and formal post-occupancy assessment. Minimum evidence: Agreed baseline, complete boundary, calibrated meters, weather and occupancy variables, change register and uncertainty method. Method: Normalise and decompose demand; compare target, baseline and outcome; test sensitivity to adjustments. Uncertainty: Show meter error, baseline model error and unmeasured changes. Safe failure: State outcome not attributable where data or change control is inadequate.

ANSWER BOUNDARY

Do not infer individual responsibility or behaviour; do not disclose tenant or person-level consumption; do not promise savings, payback or carbon outcome; do not certify energy, water or carbon compliance; do not authorise billing, targets or operational changes; do not control meters or systems.

WHY THIS IS USEFUL

Distinguish design underperformance from changed use and target investigation or future brief corrections.

AD-076

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which defects, access constraints and maintenance tasks occur more often, take longer or cost more than the design and handover evidence anticipated?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: CMMS work orders and timestamps, owner-supplied BMS alarms and runtime, vibration, temperature or current sensors only on commissioned condition-monitored assets, and panel or door-cycle events. Location and coverage: At assets and interfaces within the project scope; condition sensors are used only where the asset owner has commissioned them. Observation design: Continuous fault capture and quarterly maintenance assessment. Asset-level recurrence, downtime, task steps and cost bands. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No worker identity or productivity scoring; work-order coding, missed minor remedial works and maintenance policy changes can bias patterns.

AUTHORITATIVE SUPPORTING DATA

Owners: Estates maintenance owner, asset manager, contractor or warranty owner and responsible designer. Controlled evidence: Defects register, work orders, warranties, commissioning, O&M; requirements, change records and cost data. Precedence: Verified maintenance and contractual records over sensor alarms. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the asset owner and contract administrator where applicable.

ANALYSIS REQUIRED

Operation: Lookup + comparison + bounded diagnosis. Decision task: Failure-mode, maintainability and design-assumption comparison. Decision interval: Early occupation, end-of-defects period and annual asset assessment. Minimum evidence: Original maintainability assumption, fault history, task evidence, access condition, installation detail and operational changes. Method: Cluster recurring failures; compare evidence against product, installation, access, operation and design-interface hypotheses. Diagnosis means ranked, testable hypotheses rather than causal proof. Uncertainty: Show coding and attribution uncertainty. Safe failure: Avoid attributing blame; request inspection where evidence cannot separate causes.

ANSWER BOUNDARY

Do not infer contractor or staff competence from duration or recurrence; do not disclose personnel or commercially sensitive claim data; do not promise warranty acceptance or defect resolution; do not certify cause, liability or design negligence; do not authorise claim, payment, remediation or closure; do not control assets.

WHY THIS IS USEFUL

Correct details, information standards and access zones in future projects and prioritise proportionate remediation.

Post-occupancy evidence, design validation and lessons

AD-077

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which spatial, interface or operational hypotheses should a competent investigation test after an incident or near miss?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Authoritative incident records plus only the relevant time-aligned door, lift, alarm, environmental or asset-state data under an approved evidence hold. Location and coverage: Only at the incident location, implicated assets and a tightly bounded pre- and post-event window approved by the incident owner. Observation design: Event window with controlled pre- and post-event context. Time-aligned state summaries with provenance and access logs. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No speculative identity linkage, blame assignment, surveillance expansion or unrelated data capture; missing data, clock differences and coincidence prevent causal conclusions without competent investigation.

AUTHORITATIVE SUPPORTING DATA

Owners: Incident owner, health-and-safety lead, security or responsible person, insurers where authorised, and competent investigators. Controlled evidence: Incident and near-miss records, witness evidence, inspection, maintenance, drawings, changes and retained telemetry. Precedence: Formal investigation and competent evidence over analytic correlation. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the incident owner and competent investigator.

ANALYSIS REQUIRED

Operation: Lookup + observation + bounded diagnosis. Decision task: Event reconstruction support and design-factor hypothesis screening. Decision interval: Only when formally requested within the incident process. Minimum evidence: Verified time, location, incident description, relevant state evidence, chain of custody and investigator scope. Method: Align sources; identify plausible design-related factors and contradictory evidence without assigning cause. Diagnosis means ranked, testable hypotheses rather than causal proof. Uncertainty: Show missing data, clock error and alternative explanations. Safe failure: Return no conclusion and preserve evidence for the competent investigation.

ANSWER BOUNDARY

Do not infer fault, intent, identity or liability; do not disclose personal, security, legal or claim information outside permission; do not promise cause determination or risk elimination; do not certify safety, liability or compliance; do not authorise disciplinary, legal, insurance or remedial action; do not control systems during or after an incident.

WHY THIS IS USEFUL

Frame a competent investigation, preserve relevant evidence and decide whether a detail, interface or operating rule needs action.

AD-078

ARCHITECTS AND BUILDING DESIGNERS QUESTION

How do users describe the building's strengths, frustrations and unmet needs, and where does verified evidence support, qualify or conflict with those accounts?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Consented accessible surveys or interviews and quality-controlled environmental or use telemetry at matching broad locations. Location and coverage: At spaces covered by the consented survey, with feedback locations coded no finer than the agreed privacy threshold. Observation design: After a first settling period and at selected seasonal or mode assessments. Theme counts, distributions and representative paraphrases with minimum-group thresholds. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: Anonymity, minimum-group thresholds and no linkage between narratives and individual traces; self-selection, question wording, small groups and telemetry proxies limit generalisation.

AUTHORITATIVE SUPPORTING DATA

Owners: Client engagement lead, equality lead, data-protection owner and responsible designer. Controlled evidence: Consented feedback, consultation protocol, current use, issue register and relevant operational evidence. Precedence: Direct user evidence in context over behavioural inference; consent and privacy rules over analytic convenience. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the engagement and data owners.

ANALYSIS REQUIRED

Operation: Comparison. Decision task: Mixed-method thematic analysis and evidence triangulation. Decision interval: Planned post-occupancy assessments and after major change. Minimum evidence: Clear questions, accessible channels, respondent context collected minimally, representative telemetry and decision owner. Method: Code themes transparently; compare with evidence without treating disagreement as user error. Uncertainty: Show participation bias, minority views and proxy limits. Safe failure: Report uncertainty and seek targeted consultation rather than averaging away conflict.

ANSWER BOUNDARY

Do not infer identity, protected characteristic or motive from feedback; do not disclose raw narratives or small-group results; do not promise representativeness, satisfaction or action on every comment; do not certify design success; do not authorise issue closure or change; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Prioritise concrete changes while preserving lived experience, minority views and conflicting evidence for future briefs.

Post-occupancy evidence, design validation and lessons

AD-079

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which changes, overrides and workarounds have occupants or operators introduced, and which original design assumptions do they challenge?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: Read-only BMS override states, magnetic contacts on movable partitions and held doors, tagged furniture surveys, verified appliance submeter data and controlled work-order or change records. Location and coverage: At approved BMS points, movable elements and physically verified change locations; no person-linked logs. Observation design: Continuous state history where authorised plus quarterly physical verification. Frequency, duration and recurrence summaries by system or zone. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: No individual attribution, network-content inspection or research-content inference; an override may be legitimate, temporary or unrelated to design; undocumented changes break mapping.

AUTHORITATIVE SUPPORTING DATA

Owners: System owners, estates operations, client change-control owner and responsible designer. Controlled evidence: Approved control narrative, operating procedures, change records, work orders, user feedback and current room use. Precedence: Authorised change records and owner explanation over telemetry labels. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the relevant system owner and client sponsor.

ANALYSIS REQUIRED

Operation: Lookup + observation + bounded diagnosis. Decision task: Design-assumption drift and workaround classification. Decision interval: Seasonal post-occupancy assessment and after persistent override or repeated change. Minimum evidence: Original intent, current authorised use, state history, owner explanation and consequence evidence. Method: Classify temporary exception, commissioning issue, operational preference, changed requirement or design mismatch. Diagnosis means ranked, testable hypotheses rather than causal proof. Uncertainty: Show motive and causal uncertainty. Safe failure: Do not recommend reversing an override without owner and safety assessment.

ANSWER BOUNDARY

Do not infer misuse, intent, competence or productivity; do not disclose person-level control or access behaviour; do not promise that removing a workaround restores performance; do not certify correct control intent or safe operation; do not authorise override removal, layout reset or policy change; do not control BMS, doors or equipment.

WHY THIS IS USEFUL

Distinguish genuine learning from non-compliance and update details, controls or future assumptions accordingly.

AD-080

ARCHITECTS AND BUILDING DESIGNERS QUESTION

Which lessons are sufficiently supported and transferable to change future briefs, standards, details and evidence plans, and which remain specific to this context?

USE ONLY AFTER DEPLOYMENT, MAPPING, PERMISSION AND EVIDENCE-QUALITY GATES PASS

SAFE FAILURE: INSUFFICIENT EVIDENCE

SENSORS AND TELEMETRY

Proposed evidence: No new sensing is justified; use only quality-tagged evidence already collected for comparable decisions and contexts. Location and coverage: No new placement is assumed; use only previously verified locations that are comparable across the selected cases. Observation design: Synthesised at agreed annual or project-close learning assessments. Lesson-level evidence strength, applicability and contradiction summaries. Technical gate: Verify ID and mapped location, commissioning, applicable calibration or functional check, units/range, clock and time-zone alignment, uptime/completion, gaps, drift/outliers and raw-to-summary provenance. Privacy and validity limits: De-identify people and suppress restricted or small-group information; publication bias, changed context, weak baselines and survivorship can turn local anecdotes into false rules.

AUTHORITATIVE SUPPORTING DATA

Owners: Client design-governance owner, estates asset manager, responsible designers and discipline specialists. Controlled evidence: Approved post-occupancy reports, commissioning, defects, maintenance, incidents, change registers, feedback and design rationale. Precedence: Verified source records and competent interpretation over dashboard summaries or memorable anecdotes. Time, permission and provenance: Record the effective date of each requirement or approval, observed time of every survey or measurement, retrieved time, role-based permission, source/version and all transformations. Staleness and conflict route: Do not use superseded, expired or owner-unconfirmed evidence as unfit; keep conflicts visible and route each to the design-governance owner and client sponsor.

ANALYSIS REQUIRED

Operation: Comparison + recommendation. Decision task: Cross-evidence synthesis, lesson grading and transferability assessment. Decision interval: At project close, annual standards assessment and before a new brief adopts the lesson. Minimum evidence: Clear claim, source set, context, comparator, uncertainty, contradiction and proposed scope of application. Method: Grade as project-specific, plausible, replicated or established; test similarity before transfer. Uncertainty: Show causal limits, missing counterexamples and confidence. Safe failure: Retain as a hypothesis and define the next evidence need when generalisation is not justified.

ANSWER BOUNDARY

Do not infer universal user needs or causal design rules from one building; do not disclose confidential project, incident or user information; do not promise that a lesson will reproduce elsewhere; do not certify design standard, compliance or professional guidance; do not authorise policy or standard change; do not control doors, lifts, lighting, HVAC, alarms or other building systems.

WHY THIS IS USEFUL

Change future briefs, standards, details and evidence plans only where confidence, applicability and contextual limits are explicit.

Decision coverage and expected outputs

	CATEGORY	QUESTIONS	PRIMARY DECISION	EXPECTED OUTPUT
1	Brief, stakeholders and user requirements	001-010	Freeze, change or assign requirements	Signed requirement status and unresolved owner
2	Existing conditions, surveys and constraints	011-020	Verify the base and commission targeted survey	Survey matrix, tolerances and gap commission
3	Spatial configuration, circulation and adjacencies	021-030	Select layouts, routes and operational relationships	Option comparison, route tests and fallback plan
4	Accessibility, inclusion and universal design	031-040	Remove barriers through consultation and measured evidence	Consultation-linked barrier actions and open issues
5	Daylight, lighting, acoustics, air and comfort	041-050	Coordinate environmental quality and cross-domain trade-offs	Performance criteria, trade-off record and test plan
6	Structure, services, interfaces and maintainability	051-060	Resolve technical interfaces and future access	Coordinated interfaces, access strategy and handover tests
7	Materials, lifecycle, adaptability and carbon	061-070	Choose whole-life, low-carbon and change-ready strategies	Carbon, lifecycle, reuse and adaptability comparison
8	Post-occupancy evidence, design validation and lessons	071-080	Compare outcomes and transfer bounded learning	Post-occupancy findings, confidence and future actions

ACTION AT EACH EVIDENCE GATE

- 1
- Name the decision and owner
- 2
- Verify current authority
- 3
- Check physical coverage
- 4
- Check timing and clock alignment
- 5
- Check calibration, uptime, gaps and drift
- 6
- State method and minimum evidence
- 7
- Expose uncertainty and conflicts
- 8
- Stop safely or assign accountable action

EXPECTED DESIGN OUTPUT

A complete answer is not a recommendation alone. It should identify the controlled source, measured or assumed boundary, evidence interval, data-quality gate, method, uncertainty, conflict owner and next authorised action. Where any essential element is missing, the output should state insufficient evidence and specify the survey, consultation, owner decision or competent assessment needed.

MINIMUM DECISION RECORD

Question ID | decision owner | evidence boundary
| authoritative source | operation | uncertainty |
safe failure | accountable next action

A complete answer is not design sign-off. It is a bounded evidence statement that supports an accountable professional decision.

Rules for trustworthy use

1 VERIFY PHYSICAL TRUTH

Confirm modality, asset or survey ID, mapped position, commissioning, applicable calibration or functional check, units, range and installation context.

3 KEEP THREE TIMES

Keep effective, observed and retrieved times. A recent retrieval does not make an old source current.

5 CHECK COVERAGE AND HEALTH

Show spatial coverage, sampling, aggregation, uptime or completion, gaps, replacements, drift, outliers and excluded periods.

7 DO NOT SILENTLY MERGE CONFLICTS

Keep conflicting drawings, surveys, logs and testimony visible. Route each conflict to the named accountable owner.

9 USE BOUNDED ANALYTICS

Name the operation and method. Correlation is not cause; proxies are not the target condition; forecasts need stated assumptions.

11 FAIL SAFELY

Return insufficient evidence when an authority, permission, coverage, quality or minimum-evidence gate fails. Request the next competent action.

2 USE CURRENT AUTHORITY

Current controlled records and competent professionals take precedence. The May 2021 floor plan remains historical reference only.

4 KEEP CLOCKS ALIGNED

Record time zone, clock source, offset and drift. Do not combine event sequences when alignment is inadequate.

6 APPLY SOURCE PRECEDENCE

State which owner and controlled source decide the issue. Telemetry supports but does not override an approved requirement or competent survey.

8 PRESERVE PRIVACY

Use the least intrusive modality. Do not identify, track, profile, infer disability, inspect content or expose restricted building information.

10 SHOW UNCERTAINTY

Report tolerances, ranges, sample limits, model sensitivity, missing evidence and contexts where the result may not transfer.

12 KEEP HUMAN ACCOUNTABILITY

The service does not sign off design, certify accessibility, safety or compliance, authorise work or control building systems.

MINIMUM EVIDENCE BUNDLE

Decision owner | question ID | source precedence | effective, observed and retrieved times | survey or instrument register | mapped locations | calibration or functional checks | clock report | uptime, gaps and drift | permissions | method | uncertainty | safe failure | accountable action